

PrimeWave™ Design Environment Quick Start Guide

Version W-2024.09, September 2024



Copyright and Proprietary Information Notice

© 2024 Synopsys, Inc. This Synopsys software and all associated documentation are proprietary to Synopsys, Inc. and may only be used pursuant to the terms and conditions of a written license agreement with Synopsys, Inc. All other use, reproduction, modification, or distribution of the Synopsys software or the associated documentation is strictly prohibited.

Destination Control Statement

All technical data contained in this publication is subject to the export control laws of the United States of America. Disclosure to nationals of other countries contrary to United States law is prohibited. It is the reader's responsibility to determine the applicable regulations and to comply with them.

Disclaimer

SYNOPSYS, INC., AND ITS LICENSORS MAKE NO WARRANTY OF ANY KIND, EXPRESS OR IMPLIED, WITH REGARD TO THIS MATERIAL, INCLUDING, BUT NOT LIMITED TO, THE IMPLIED WARRANTIES OF MERCHANTABILITY AND FITNESS FOR A PARTICULAR PURPOSE.

Trademarks

Synopsys and certain Synopsys product names are trademarks of Synopsys, as set forth at <https://www.synopsys.com/company/legal/trademarks-brands.html>.

All other product or company names may be trademarks of their respective owners.

Free and Open-Source Licensing Notices

If applicable, Free and Open-Source Software (FOSS) licensing notices are available in the product installation.

Third-Party Links

Any links to third-party websites included in this document are for your convenience only. Synopsys does not endorse and is not responsible for such websites and their practices, including privacy practices, availability, and content.

www.synopsys.com

Contents

New in This Release	5
Related Products, Publications, and Trademarks	5
Conventions	6
Customer Support	7
Statement on Inclusivity and Diversity	8
1. Getting Started With Netlist-Based Flow in the PrimeWave Design Environment	9
Before You Start	10
Specify a File Text Editor	11
Starting the Software	12
Importing a Netlist	14
Re-Importing a Netlist	19
PrimeWave Design Environment Software User Interface	22
Elements of the PrimeWave Design Environment User Interface	23
Data Transfer from a Source File to the PrimeWave Design Environment UI and Include Files	24
Context Menus	25
Shortcut Keys	25
Exiting the PrimeWave Design Environment	26
Examining the Design	26
Adding Specifications to the Testbench	28
Running a Simulation and Viewing the Results	29
Rerunning a Simulation and Viewing History Points	33
Importing a Post-Layout Version of the Circuit Under Test	35
Running an Advanced Analysis: Monte Carlo Analysis	41
Adding a New Testbench and Viewing the Model Files	42
Setting Up and Running the Monte Carlo Analysis	45
Viewing the Results	47
Performing Sensitivity Analysis	52
Managing Designs with the Design Manager	53

Contents

Opening the Design Manager	53
Opening Existing Test Suites	54
Deleting Test Suites	54
Opening Results from Previous Simulations	55
Viewing Designs	55

2. Getting Started With Schematic-Based Flow in the PrimeWave Design Environment	56
Quick Start Instructions	57
Advantages of Standalone PrimeWave	58

About the PrimeWave Design Environment Quick Start Guide

The *PrimeWave Design Environment Quick Start Guide* introduces you to the netlist based and schematic based PrimeWave Design Environment, which uses input and lets you set up testbenches and run simulations with Synopsys circuit simulators and view results with the WaveView tool. For detailed information about the PrimeWave Design Environment features not described in this guide, see the *PrimeWave™ Design Environment User Guide*.

This preface includes the following sections:

- [New in This Release](#)
- [Related Products, Publications, and Trademarks](#)
- [Conventions](#)
- [Customer Support](#)
- [Statement on Inclusivity and Diversity](#)

New in This Release

Information about new features, enhancements, and changes, known limitations, and resolved Synopsys Technical Action Requests (STARs) is available in the PrimeWave Design Environment Release Notes on the SolvNetPlus site.

Related Products, Publications, and Trademarks

For additional information about the PrimeWave™ Design Environment tool, see the documentation on the Synopsys SolvNetPlus support site at the following address:

<https://solvnetplus.synopsys.com>

You might also want to see the documentation for the following related Synopsys products:

Synopsys® PrimeWave™ Design Environment

Synopsys® Custom Compiler™

PrimeSim™ Continuum™

PrimeSim™

- PrimeSim™ Pro™
- PrimeSim™ SPICE™
- PrimeSim™ HSPICE®
- PrimeSim™ XA™
- PrimeSim™ CCK™
- PrimeSim™ MOSRA™
- VCS® PrimeSim™ AMS™
- Synopsys® FineSim®
- Synopsys® HSPICE®
- Synopsys® VCS®
- Synopsys® Verdi®
- Cadence® Spectre® Circuit Simulator
- MathWorks® MATLAB

Conventions

The following conventions are used in Synopsys documentation.

Convention	Description
Courier	Indicates syntax, such as <code>write_file</code> .
<i>Courier italic</i>	Indicates a user-defined value in syntax, such as <code>write_file design_list</code>
Courier bold	Indicates user input—text you type verbatim—in examples, such as <code>prompt> write_file top</code>
Purple	• Within an example, indicates information of special interest. • Within a command-syntax section, indicates a default, such as <code>include_enclosing = true false</code>
[]	Denotes optional arguments in syntax, such as <code>write_file [-format fmt]</code>
...	Indicates that arguments can be repeated as many times as needed, such as <code>pin1 pin2 ... pinN</code> .

Convention	Description
	Indicates a choice among alternatives, such as low medium high
\	Indicates a continuation of a command line.
/	Indicates levels of directory structure.
Bold	Indicates a graphical user interface (GUI) element that has an action associated with it.
Edit > Copy	Indicates a path to a menu command, such as opening the Edit menu and choosing Copy .
Ctrl+C	Indicates a keyboard combination, such as holding down the Ctrl key and pressing C.

Customer Support

Customer support is available through SolvNetPlus.

Accessing SolvNetPlus

The SolvNetPlus site includes a knowledge base of technical articles and answers to frequently asked questions about Synopsys tools. The SolvNetPlus site also gives you access to a wide range of Synopsys online services including software downloads, documentation, and technical support.

To access the SolvNetPlus site, go to the following address:

<https://solvnetplus.synopsys.com>

If prompted, enter your user name and password. If you do not have a Synopsys user name and password, follow the instructions to sign up for an account.

If you need help using the SolvNetPlus site, click REGISTRATION HELP in the top-right menu bar.

Contacting Customer Support

To contact Customer Support, go to <https://solvnetplus.synopsys.com>.

Statement on Inclusivity and Diversity

Synopsys is committed to creating an inclusive environment where every employee, customer, and partner feels welcomed. We are reviewing and removing exclusionary language from our products and supporting customer-facing collateral. Our effort also includes internal initiatives to remove biased language from our engineering and working environment, including terms that are embedded in our software and IPs. At the same time, we are working to ensure that our web content and software applications are usable to people of varying abilities. You may still find examples of non-inclusive language in our software or documentation as our IPs implement industry-standard specifications that are currently under review to remove exclusionary language.

1

Getting Started With Netlist-Based Flow in the PrimeWave Design Environment

This chapter explains how to prepare your environment to use the netlist-based flow in the PrimeWave Design Environment, import a netlist, add output expressions and simulator-specific commands, run a simulation, and view the results. A tutorial format is used to illustrate these tasks.

The PrimeWave Design Environment netlist-based flow provides a fully featured simulation environment and shares many features found in the schematic-driven PrimeWave Design Environment tool. Simulators are used to verify analog and mixed-signal designs. In the PrimeWave Design Environment, you can:

- set up or modify simulator control statements (such as analyses, models, and options)
- import an existing netlist into the user interface
- run simulations and analyze the results
- perform parametric sweeps, corners analysis, and Monte Carlo analysis
- post-process results using the waveform viewer
- use the Results Analyzer for plotting and expression building

After completing the following tutorial, you are encouraged to explore the PrimeWave Design Environment features that are not described in this quick start guide. For detailed information about all the PrimeWave Design Environment features, including how to invoke the schematic-based flow from the Custom Compiler tool, see the *PrimeWave Design Environment User Guide*. References to that guide have been provided throughout this tutorial to help you pursue more advanced information on specific topics.

This chapter contains the following topics:

- [Before You Start](#)
- [Starting the Software](#)
- [Importing a Netlist](#)

- [PrimeWave Design Environment Software User Interface](#)
- [Examining the Design](#)
- [Adding Specifications to the Testbench](#)
- [Running a Simulation and Viewing the Results](#)
- [Importing a Post-Layout Version of the Circuit Under Test](#)
- [Running an Advanced Analysis: Monte Carlo Analysis](#)
- [Managing Designs with the Design Manager](#)

Before You Start

Make sure you have the following:

Item	Description	Location
One or more of the following simulators, depending on availability: • PrimeSim HSPICE • PrimeSim SPICE • PrimeSim XA	Software	Installed on the server
<code>PrimeWave_Netlist_Flow_Lab</code>	Lab data folder	Name of the directory containing the tutorial lab data
<code>spice_lib</code>	Directory containing include files for the design	The <code>PrimeWave_Netlist_Flow_Lab</code> directory
<code>saved_specs</code>	Directory containing saved state files to be loaded	The <code>PrimeWave_Netlist_Flow_Lab</code> directory
<code>models</code>	Directory containing model library file	The <code>PrimeWave_Netlist_Flow_Lab</code> directory
<code>lfsr_v1.spi</code>	Testbench netlist suitable for simulation with PrimeSim HSPICE, PrimeSim SPICE, and PrimeSim XA	The <code>PrimeWave_Netlist_Flow_Lab</code> directory

Item	Description	Location
lfsr_v2.spi	Testbench netlist suitable for simulation with PrimeSim HSPICE, PrimeSim SPICE, and PrimeSim XA	The PrimeWave_Netlist_Flow_Lab directory
opampmc.sp	Testbench netlist suitable for simulation with PrimeSim HSPICE, PrimeSim SPICE, and PrimeSim XA	The PrimeWave_Netlist_Flow_Lab directory
.sae.tcl	Initialization file for the PrimeWave Design Environment	The PrimeWave_Netlist_Flow_Lab directory. This is a hidden file. To view the file, use the <code>ls -la</code> command in the Linux terminal.

The PrimeWave_Netlist_Flow_Lab data folder is contained in a compressed archive included with the software installation and is located here:
`<product_installation_directory>/doc/tutorials/PrimeWave_NL.tar.gz`.

1. Uncompress and extract this archive in a directory where you have read and write privileges by typing the following at the command line:

```
% tar xzf <product_installation_directory>/doc/tutorials/  
PrimeWave_NL.tar.gz
```

2. Descend into the PrimeWave_Netlist_Flow_Lab directory:

```
% cd PrimeWave_Netlist_Flow_Lab
```

Caution:

Avoid running the PrimeWave Design Environment directly in your home directory (the default). Running lots of simulations in `$HOME` can create large amounts of data, which can then exceed the allotted disk quota.

Specify a File Text Editor

To specify an external file text editor, which is used to edit netlists in the PrimeWave Design Environment, set the `EDITOR` environment variable before you start the PrimeWave Design Environment.

- For csh, type the following at the command line (or add it to your `.cshrc` startup file):

```
% setenv EDITOR vi
```

- For bash, type the following at the command line before you start (or add it to your `.bashrc` startup file):

```
% export EDITOR=vi
```

Alternatively, when you start the PrimeWave Design Environment, you can choose **Options > General > Editors** from the **Home** page to set the text editor.

Starting the Software

This section describes how to start the PrimeWave Design Environment.

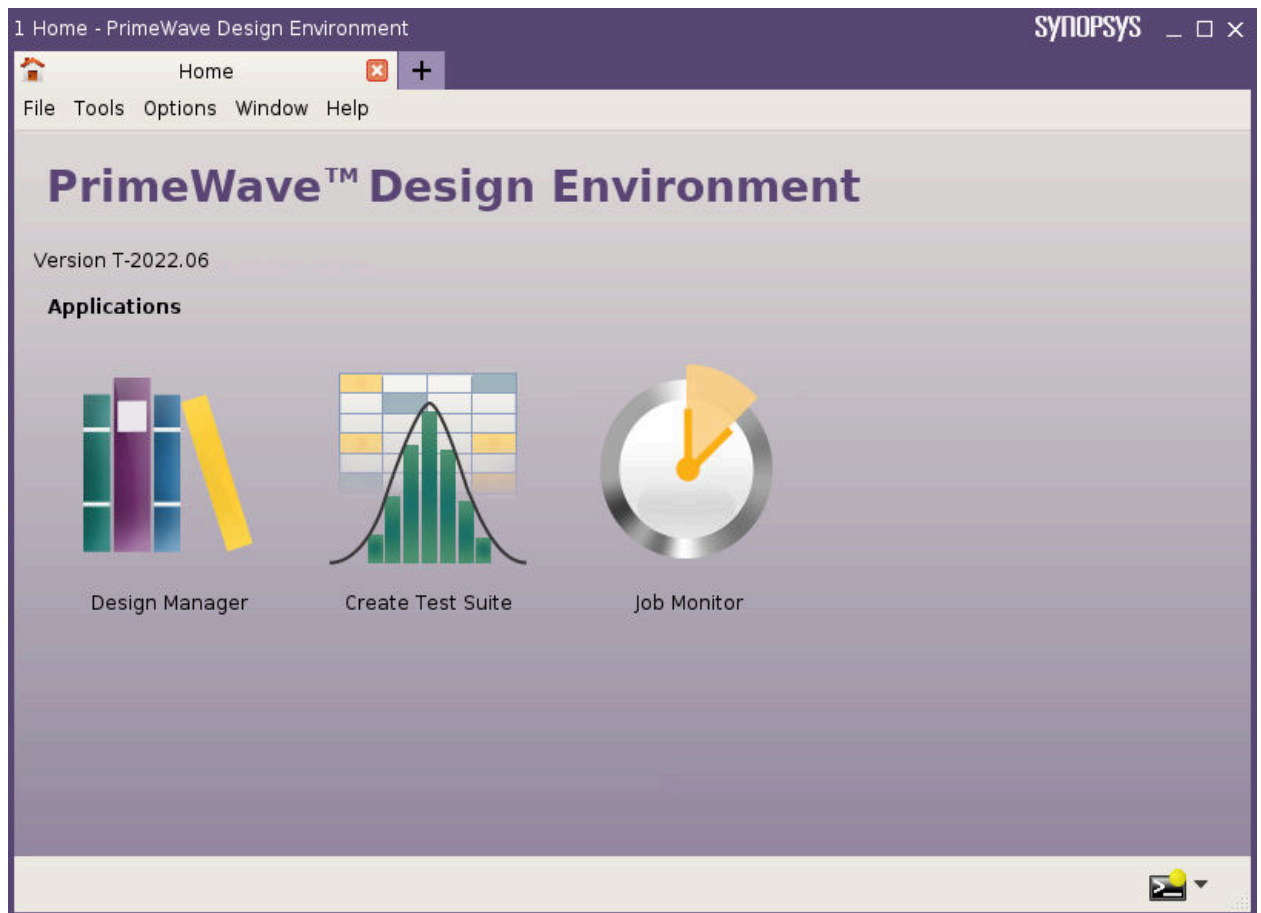
Caution:

Before you invoke the PrimeWave Design Environment, be sure to specify an editor to edit netlists. See [Specify a File Text Editor](#) for more information.

1. To invoke the PrimeWave Design Environment, enter the following at the command line:

```
% primewave &
```

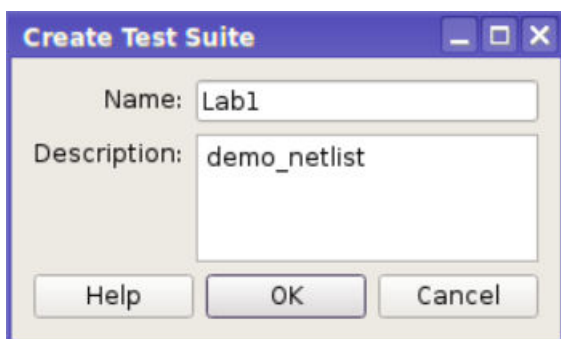
The PrimeWave Design Environment **Home** tab appears.



You are encouraged to explore the documentation and learn about the menu options that are not discussed in this manual. All PrimeWave Design Environment windows contain a **Help** item that opens a new tab with context-sensitive help, including a search feature. The *PrimeWave Design Environment User Guide* provides detailed information about all the PrimeWave Design Environment features.

2. A test suite is a single location where you can gather multiple testbenches together for simulation.

To create a new test suite, click **Create Test Suite**  on the **Home** tab. The **Create Test Suite** dialog box opens.



3. In the **Create Test Suite** dialog, enter **Lab1** for **Name**. The **Description** field is optional.
4. Click **OK** to commit the name and dismiss the dialog box.

The PrimeWave Design Environment opens in a new tab, as well as the **Import Text into New Testbench** dialog box. Continue to [Importing a Netlist](#) for instructions about how to import a netlist using the simulator of your choice.

Importing a Netlist

The new test suite created in [Starting the Software](#) does not contain any testbenches, so the **Import Text into New Testbench** dialog box automatically appears after you create the test suite.

Each testbench in a test suite can use different simulators, different setups, different designs, or a different level of abstraction (behavioral versus structural) of the same design or different designs. Features in the netlist not supported by the selected simulator are ignored.

The netlist-based PrimeWave Design Environment flow includes the ability to automatically populate the **Corners** dialog box from `.alter` blocks found in the source netlist.

For a successful import and simulation, keep in mind the following:

- Only `.param`, `.lib`, and `.temp` statements are supported in the `.alter` blocks. However, temperature sweeps (for example, `.temp 25 50 75`) in mapping to corners are not supported.
- Other statements found in `.alter` blocks are ignored and not used.

Import a SPICE netlist for simulation using one of the following PrimeSim Continuum simulators:

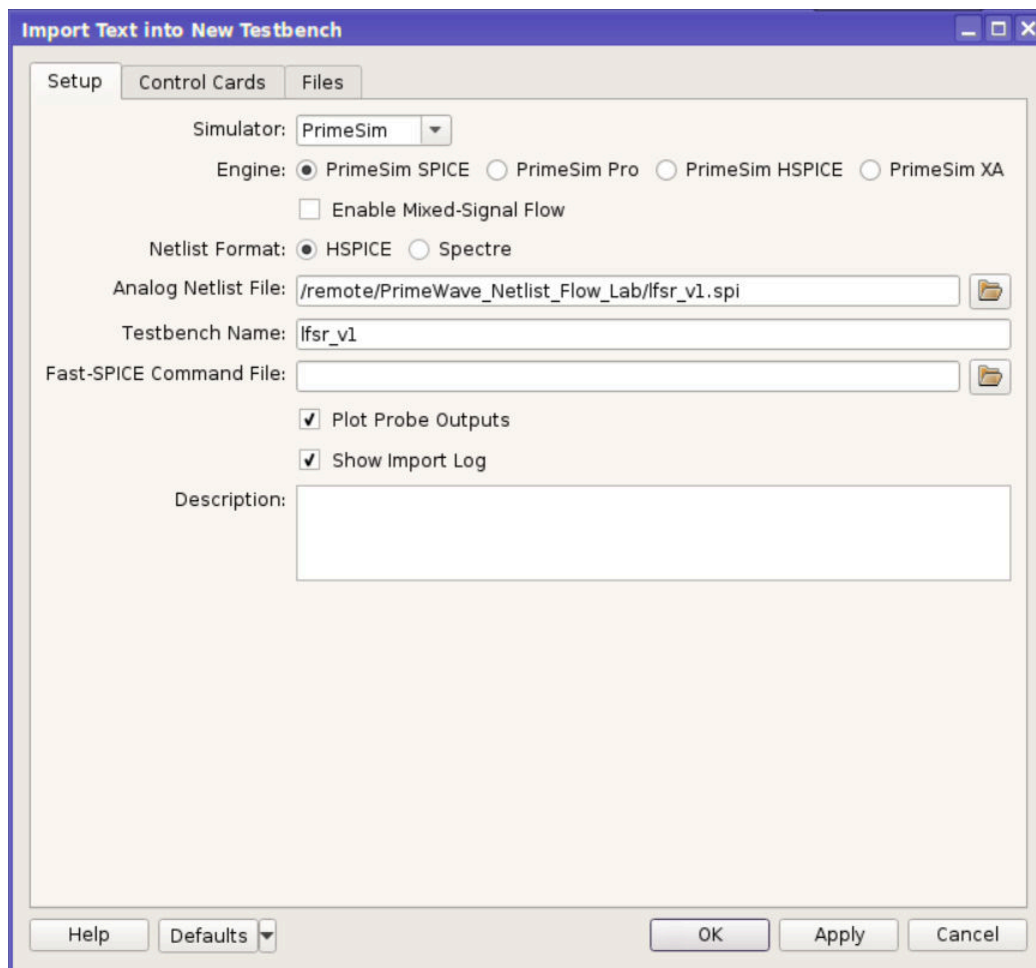
- PrimeSim Pro
- PrimeSim SPICE
- PrimeSim XA

Note:

VCS PrimeSim AMS and FineSim VCS are mixed-signal simulation integrations that support analog-centric flows and are similar to the analog integrations but also support digital data. For further information about these integrations, see Setting Mixed-Signal Simulator Options in the *PrimeWave Design Environment User Guide* or contact your Synopsys representative.

To use the netlist-based PrimeWave Design Environment with PrimeSim Continuum simulators:

1. In the **Import Text into New Testbench** dialog box, choose **PrimeSim** for the **Simulator** and **PrimeSim SPICE** for the **Engine**.



2. Browse to the **lfsr_v1.spi** file for the **Analog Netlist File**.
3. The tool automatically creates a **Testbench Name** for you. You can change the testbench name as needed.

The netlist-based PrimeWave Design Environment supports multiple testbenches from within a test suite. Each testbench must have a unique name; if the name in the **Testbench Name** field has already been used, the field is highlighted in a red box until you change it to a unique name.

4. For this tutorial, a **Fast-SPICE Command File** is not necessary.

5. **Plot Probe Outputs** is enabled by default and means that all probed items written to the PrimeWave Design Environment outputs table will have the **Plot Waveforms/Evaluate** box checked. Disabling this option results in all probed items written to the PrimeWave Design Environment outputs table having **Plot Waveforms/Evaluate** unchecked.
6. Enable **Show Import Log** to open a log file in the Text Viewer at the end of the import process, showing the details of the netlist import.
7. Switch to the **Control Cards** tab. This tab consists of different sections of the netlist that you can choose to import or discard.
8. Select **Discard** for **Alters**. Alters will be reimported in the next task, [Re-Importing a Netlist](#).

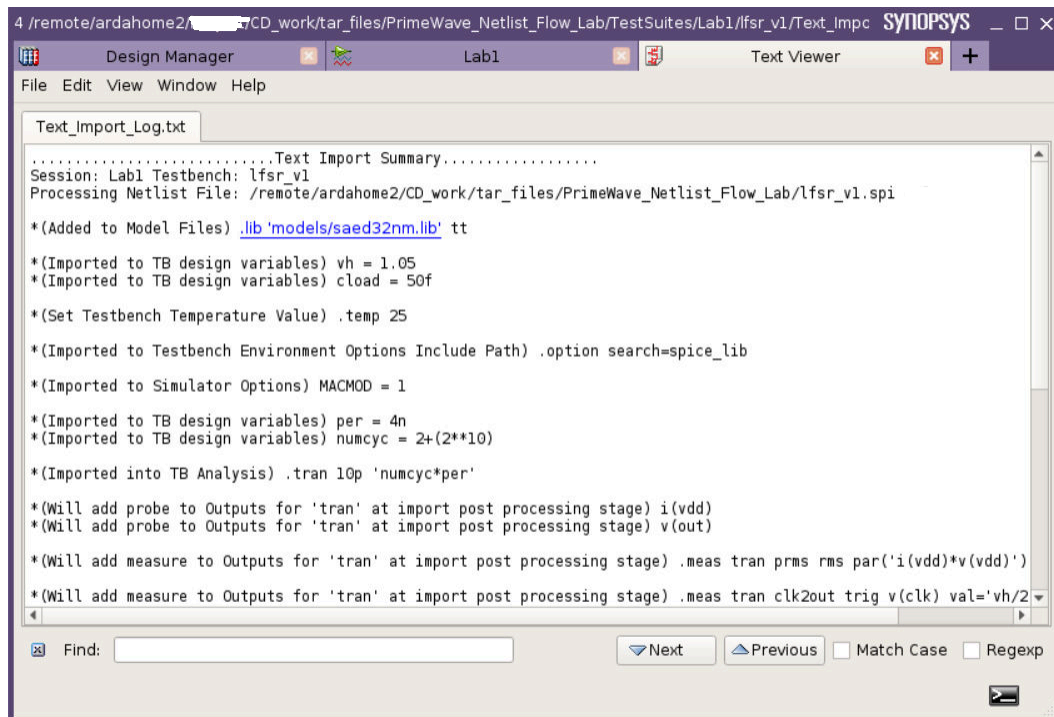


9. Switch to the **Files** tab. Any other additional files that are required to include or parse can be pointed to using the file browser. Wildcard patterns are supported.
10. Click **OK** to execute the import process.

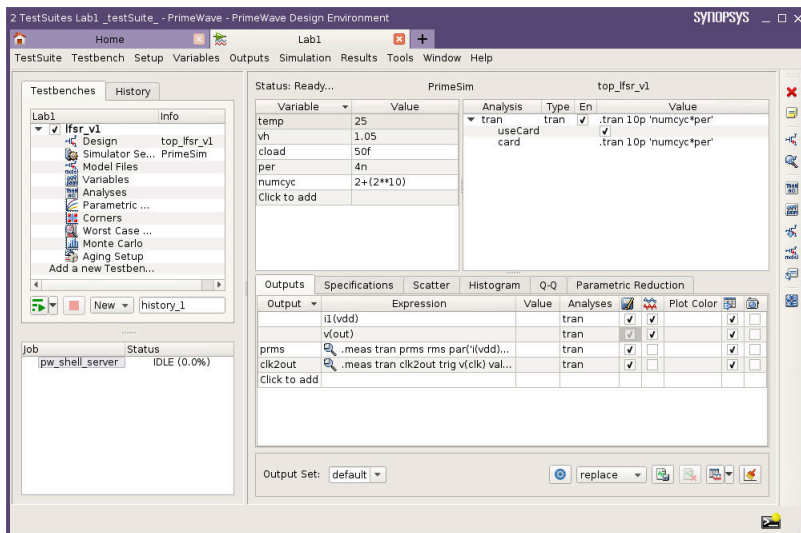
Note:

After clicking **Apply** or **OK**, you can interrupt the import process, if necessary, by pressing Ctrl+C.

When the **Show Import Log** option is enabled, the Text Viewer opens first in a new tab, showing the import netlist logs.



11. Switch to the tab labeled with the test suite name, **Lab1**. This is the main PrimeWave Design Environment window.

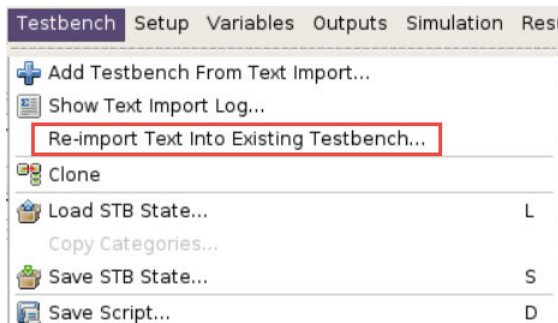


Re-Importing a Netlist

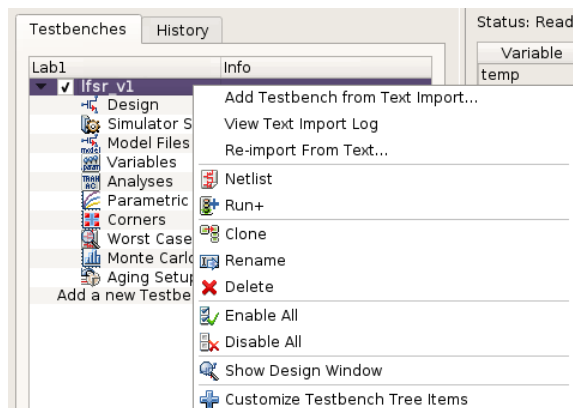
If the original imported SPICE netlist changes after you complete one or more set-up steps in the PrimeWave Design Environment, you can re-import it to update the structural netlist without losing the current PrimeWave Design Environment setup.

You can initiate the re-import using one of two methods:

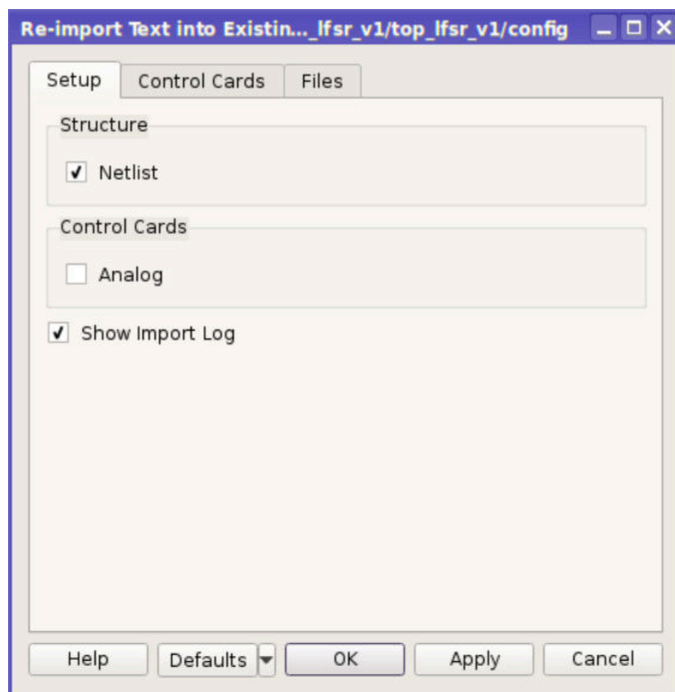
1. Choose **Testbench > Re-import Text into Existing Testbench**.



2. Alternatively, right-click on the selected testbench and choose **Re-import From Text** from the menu that appears.



The **Re-import Text into Existing Testbench** dialog box appears.



3. Make sure **Analog** is unchecked in the **Setup** tab **Control Cards** section.

Caution:

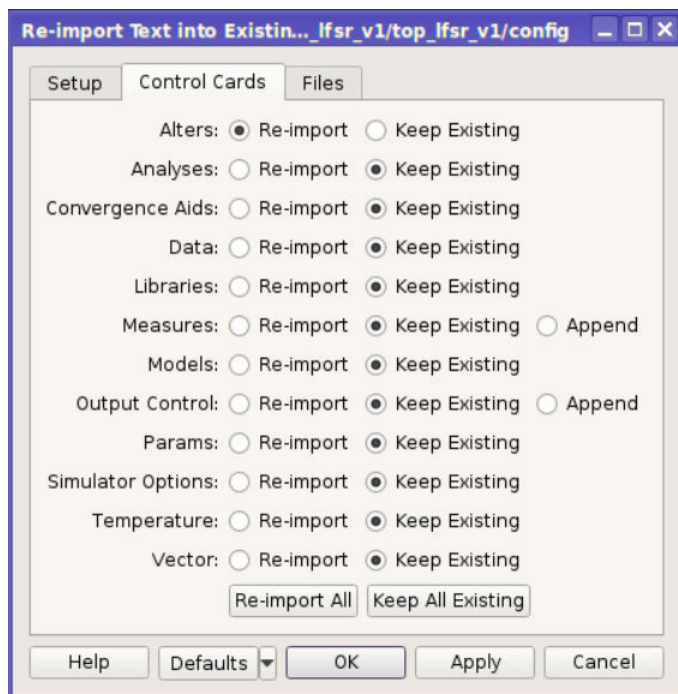
Unchecking the default **Analog** option is necessary to avoid re-importing the contents for all categories listed in the **Control Cards** tab and overwriting the data already in the PrimeWave Design Environment.

4. Switch to the **Control Cards** tab. The **Control Cards** provide finer control for specific analog netlist categories.

The list of options in the **Control Cards** tab could change depending on the simulator.

For the category **Alters**, select **Re-import**. For the remaining categories, choose **Keep Existing**.

For the **Measures** and **Output Control** categories you can choose to append these results when running simulations in Append mode. For details on Append mode, see *Running Simulations in Append Mode in the PrimeWave Design Environment User Guide*.



5. Click **OK** to import the updated structural netlist from the original file.

Note:

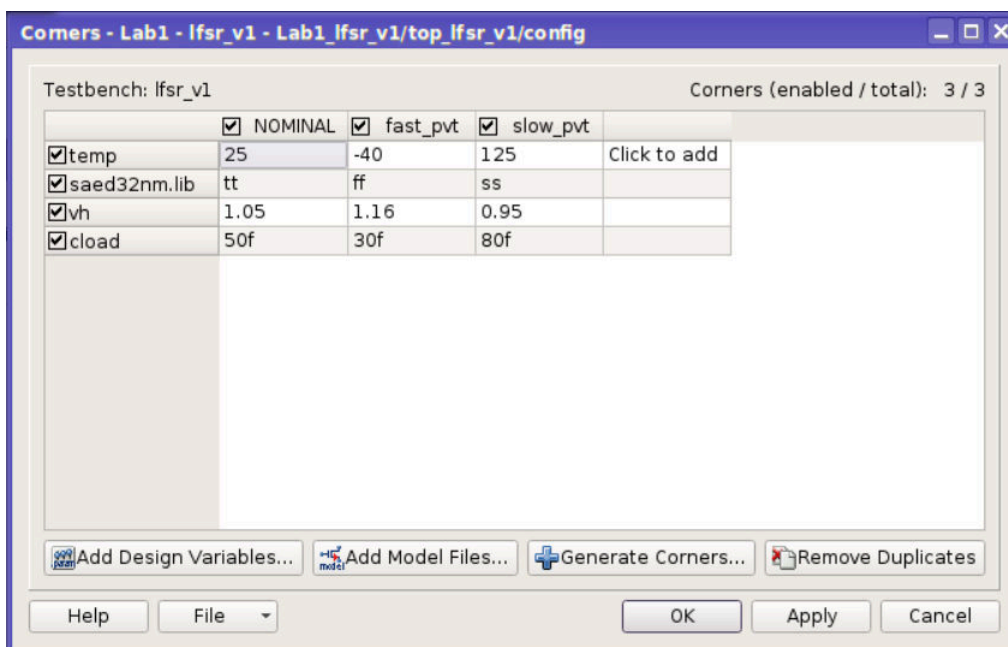
Cloned testbenches share the same design text netlists. If the originally imported netlist is changed, when you re-import it into either the original or any of the cloned testbenches, all clones and the original PrimeWave Design Environment testbenches are updated with the new netlist.

After clicking **Apply** or **OK**, you can interrupt the import process, if necessary, by pressing Ctrl+C.

In this tutorial, re-importing the netlist adds 3 corners to the testbench.

Analysis	Type	En	Value
corners		☒	Total:3, enabled: 3
▶ tran	tran	☒	.tran 10p 'numcyc*per'

- Choose **Tools > Corners** to open the **Corners** setup dialog box and review the corners.



See Also

- [Setting Up Corner Analyses](#)

PrimeWave Design Environment Software User Interface

After importing a netlist, take some time to explore the main features of the PrimeWave Design Environment. The following sections provide an overview of the PrimeWave Design Environment user interface.

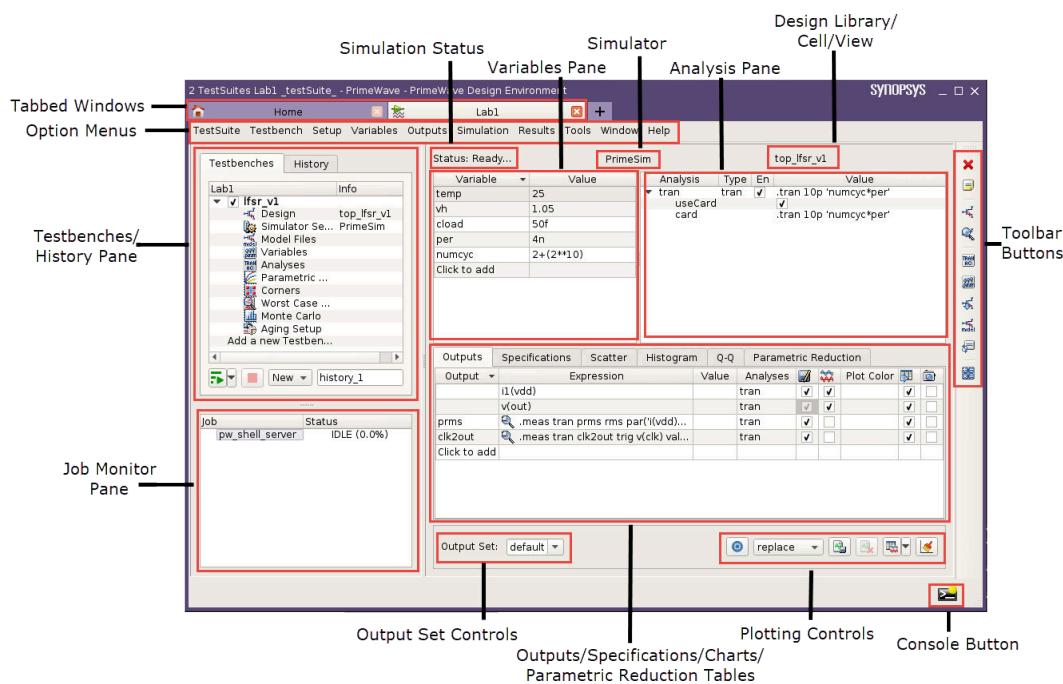
- [Elements of the PrimeWave Design Environment User Interface](#)
- [Data Transfer from a Source File to the PrimeWave Design Environment UI and Include Files](#)
- [Context Menus](#)

- [Shortcut Keys](#)
- [Exiting the PrimeWave Design Environment](#)

Elements of the PrimeWave Design Environment User Interface

Figure 1 shows the elements of the PrimeWave Design Environment user interface.

Figure 1 The PrimeWave Design Environment User Interface



The PrimeWave Design Environment allows you to arrange tabbed windows in any order, as well as drag a tab off to a standalone window or drag a window on as a tab. Figure 1 shows two tabbed windows, **Lab1** and the **Home** tab.

PrimeWave Design Environment window menus include **TestSuite**, **Testbench**, **Setup**, **Variables**, **Outputs**, **Simulation**, **Results**, **Tools**, **Window**, and **Help**. Experiment with these menus and, for more details, refer to the built-in help accessible from the **Help** menu.

On the left in Figure 1 is the **Testbenches/History** pane. The netlist-driven PrimeWave Design Environment flow operates in what is known as the multiple testbench (MTB) mode. In Figure 1, just one run-enabled testbench is shown. Each netlist imported is a unique testbench and part of a shared test suite. Under each netlist, there are several

basic entries related to setup/analyses for easy access. Double-click on these entries to directly launch corresponding tool dialog boxes.

In the upper middle is the **Variables** pane. Parameter names and values found in the source file are written to this pane.

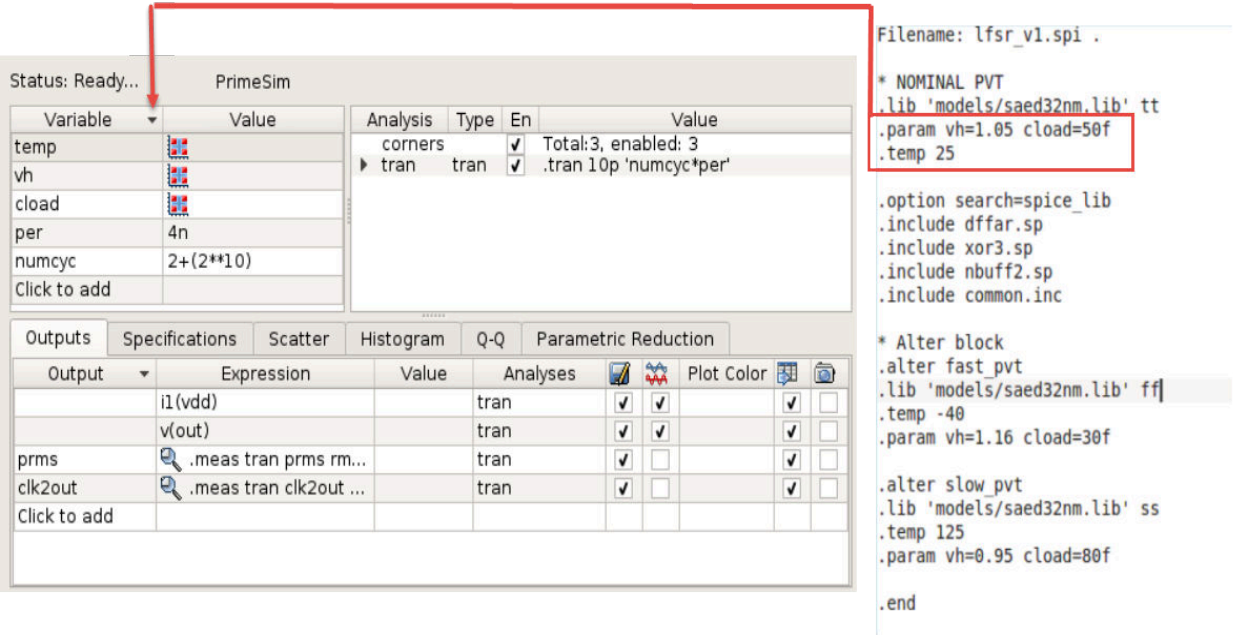
In the upper right is the **Analysis** pane. Analysis statements found in the source file are written to this pane.

In the lower right is the **Outputs/Specifications/Charts/Parametric Reduction** pane. In general, waveform probe statements and PrimeSim SPICE `.measure` statements found in the source file are written to this area.

Data Transfer from a Source File to the PrimeWave Design Environment UI and Include Files

Figure 2 illustrates how some of the text from the `lfsr_v1.spi` file is transferred to the PrimeWave Design Environment user interface.

Figure 2 Data Transfer from Source File to PrimeWave Design Environment User Interface



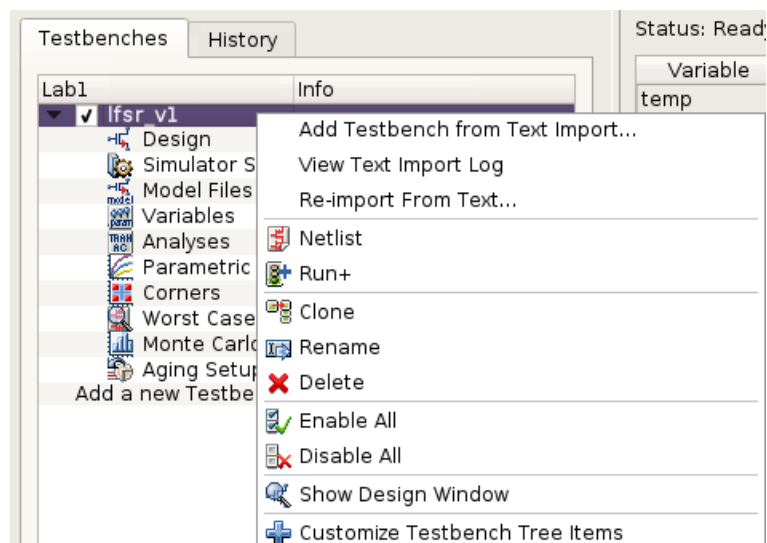
PrimeSim SPICE library file statements, typically used to define semiconductor device models, are written in to the PrimeWave Design Environment **Model Files** dialog box. Choose **Setup > Model Files** to examine this dialog box.

Text from the `lfsr_v1.spi` file that does not have an equivalent home in the graphical areas of PrimeWave Design Environment is written into text files that are included in the final netlist. Choose **Setup > Include Files** to examine the include files.

Context Menus

Many of the objects shown in [Figure 1](#) can be selected and modified using a single button click. The PrimeWave Design Environment also provides a context-sensitive menu in these panes. Right-click any object to see all the available actions you can take.

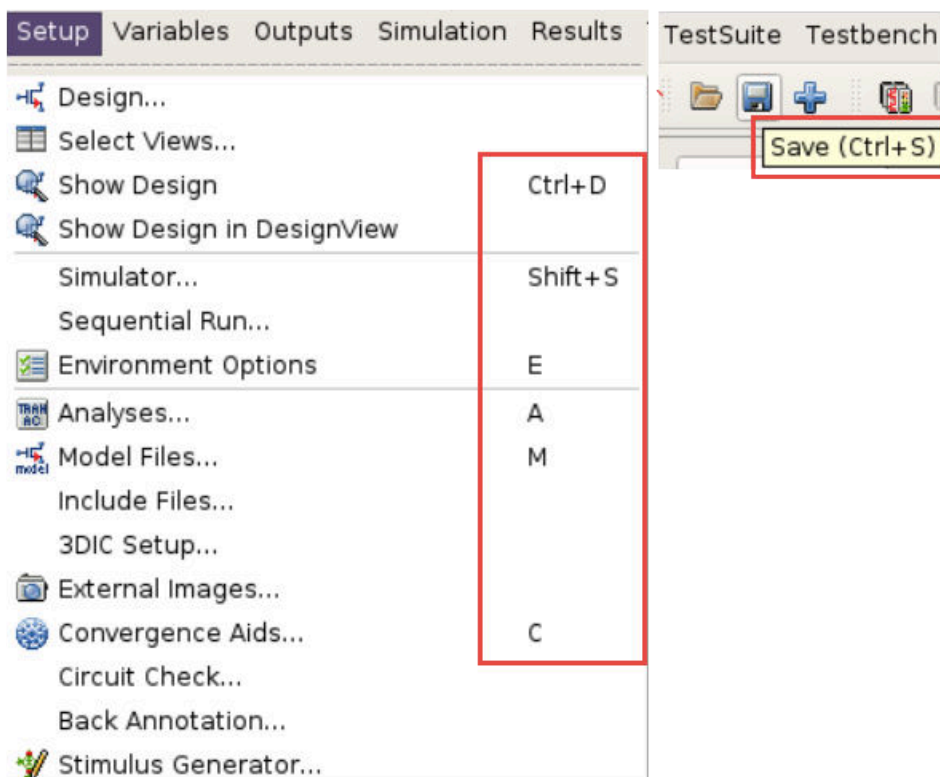
The figure below shows the menu available when right-clicking on the selected testbench.



Shortcut Keys

The PrimeWave Design Environment provides shortcut keys (also referred to as key bindings) for common actions. Shortcut key information for the currently active binding set is provided in the menus and in tooltips that come up when you hover over an icon in the PrimeWave Design Environment window. A complete list of shortcut keys is provided in PrimeWave Design Environment Shortcut Keys in the *PrimeWave Design Environment User Guide*.

For more information about customizing the binding set, see the *Custom Compiler Environment User Guide*.



Exiting the PrimeWave Design Environment

When you are ready to exit the PrimeWave Design Environment completely:

- Click the **X** in the upper-right corner of the main window; or
- Press Ctrl+W multiple times, until the **Exit - PrimeWave Design Environment** dialog box appears. Press Return or click **Exit** in the dialog box to terminate the session.

If you have any unsaved changes, press Ctrl+W while in the main PrimeWave Design Environment window to open the **Close Session** dialog box. You can discard or save your changes, or click **Cancel** to return to the main PrimeWave Design Environment window.

Examining the Design

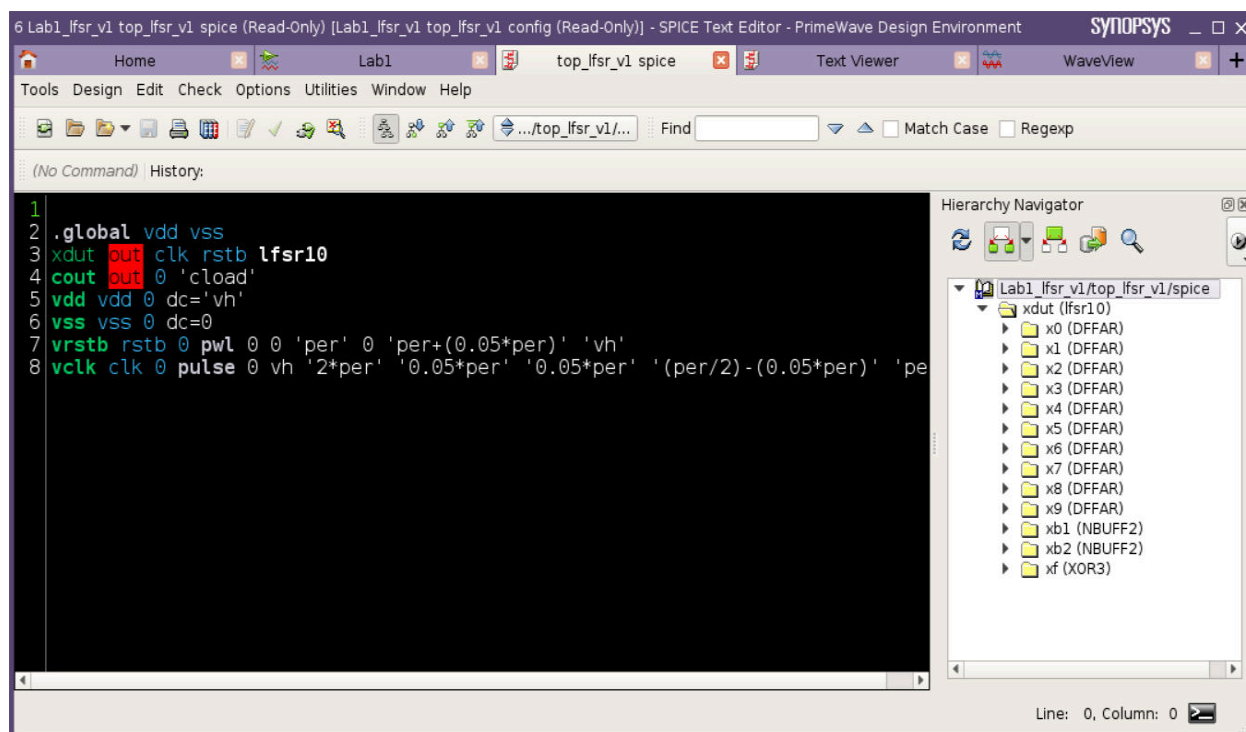
This section explores the structural design in PrimeWave Design Environment.

1. In the main PrimeWave Design Environment window, choose **Setup > Show Design** or

click **Show Design**  on the right edge of the window. Alternatively, right-click on a testbench in the **Testbenches** tab and choose **Show Design Window**.

The top-level structural design appears in a new SPICE text editor window, shown below. The window menus and icons have changed to reflect the fact that you are viewing a design.

In addition to the hierarchy navigation features available in the SPICE text editor menus, the PrimeWave Design Environment displays the Hierarchy Navigator assistant, which shows the hierarchy tree. Detailed information about the Hierarchy Navigator is provided in the *Custom Compiler Environment User Guide*.

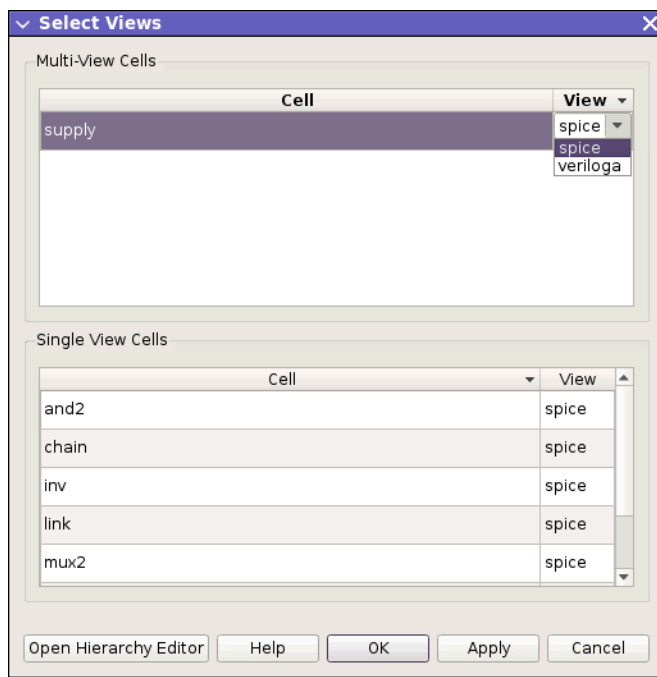


2. Double-click the `xdut` instance name text to descend into the chain subcircuit definition. You can double-click again on any of the `dffar` instances to descend into the linked subcircuit definition. Hierarchy traversal actions are accessible from the **Design > Hierarchy** menu, the **Hierarchy** toolbar icons, or the Hierarchy Navigator.

3. Choose **Design > Hierarchy > Return to Top** or click **Return to Top**  to go back to the top level.

4. For netlists where certain cells or subcircuits have more than one view, the **Select Views** dialog box allows you to select which view to use during simulation.

From the main PrimeWave Design Environment window, you can choose **Setup > Select Views** to open the **Select Views** dialog box and select which view to include in the simulation.



The **Select Views** dialog box has two panels. The lower panel (**Single View Cells**) lists each cell or subcircuit in the design that has only one view. In this panel you can import additional views for any cell in this list.

The upper panel (**Multi-View Cells**) lists all cells and subcircuits with two or more views. In this panel you can select which of the views to use for a given cell.

After applying changes, the navigation of the design uses the newly selected view.

Adding Specifications to the Testbench

Specifications are used to determine at which point a measurement is a violation. In this task, you set up basic goals for the two imported PrimeSim SPICE `.measure` expressions.

To add a specification to the lfsr_v1 testbench:

- In the **Outputs** pane, click the **Specifications** tab.

Outputs	Specifications	Scatter	Histogram	Q-Q	Parametric Reduction	
En	Measurement	Goal Type	Goal/ Range Low	Range High	Marginal/ Marginal Low	Marginal High
☒	prms	>		N/A	0%	N/A
☒	clk2out	>		N/A	0%	N/A
	Click to add					

- Set the **Goal Type** and **Goal/Range Low** for the prms measure to be < 0.5m.
- Set the **Goal Type** and **Goal/Range Low** for the clk2out measurement to be < 0.6n. The **Specifications** table should look like as follows.

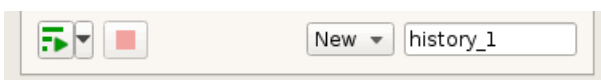
Outputs	Specifications	Scatter	Histogram	Q-Q	Parametric Reduction	
En	Measurement	Goal Type	Goal/ Range Low	Range High	Marginal/ Marginal Low	Marginal High
☒	prms	<	0.5	N/A	0%	N/A
☒	clk2out	<	0.6	N/A	0%	N/A
	Click to add					

Running a Simulation and Viewing the Results

This section describes how to netlist the design and run a single (or *nominal*) simulation, followed by review of the analog waveform and scalar results. All the functions of the WaveView tool are integrated in the PrimeWave Design Environment.

To run a simulation and view the results:

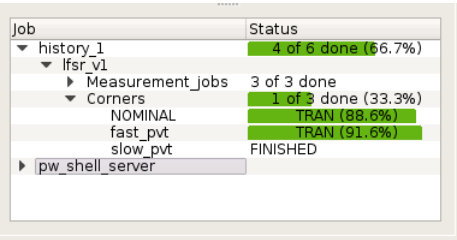
- Click the green netlist and run button  to run the simulations in history_1, which is the default name given to the first history_point. You can edit this name as per context. The drop-down beside the history name helps toggle between new and current history points.



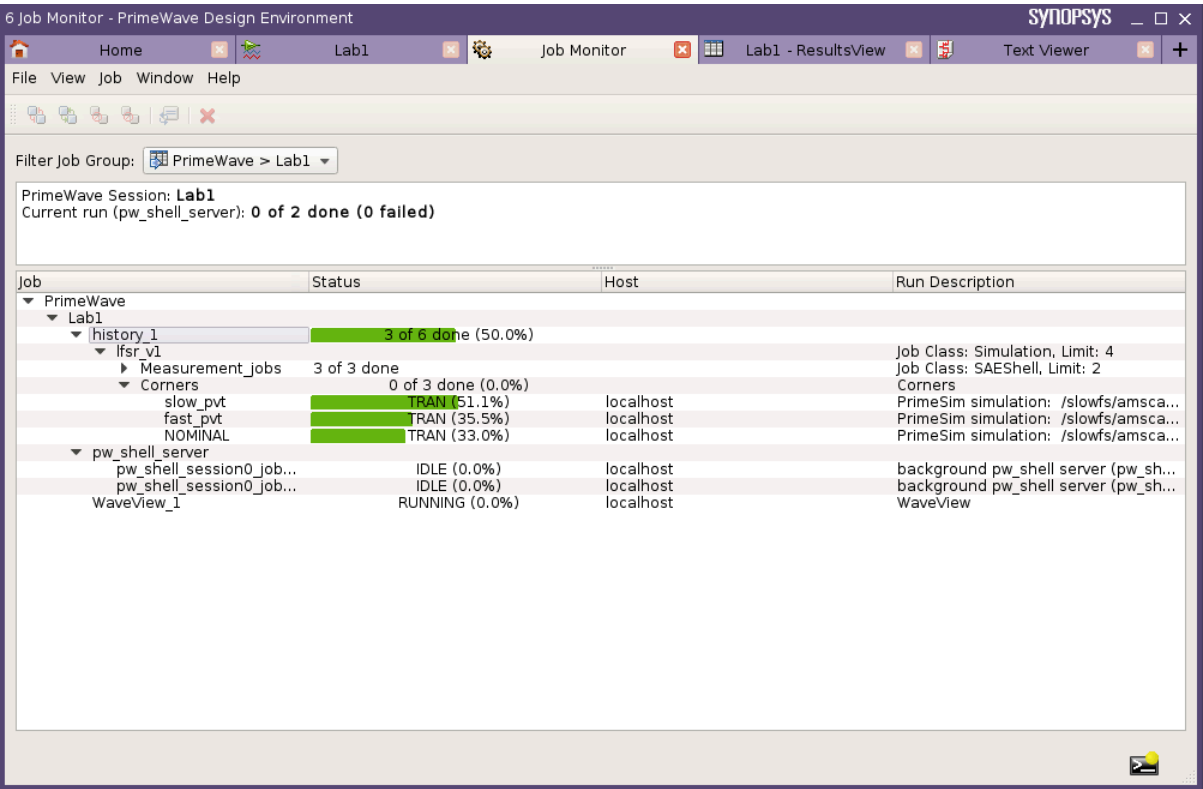
Note:

The contents of the toolbar can be controlled using **Window > Toolbars**.

By default, the job status is displayed in the bottom-left panel, showing the real-time status of the simulation.



To view more details, you can open the Job Monitor in a new tab using **Tools > Show Simulation Jobs**.



Following the successful simulation, the ResultsView tool appears in another tab showing results of the scalar calculations.

Chapter 1: Getting Started With Netlist-Based Flow in the PrimeWave Design Environment

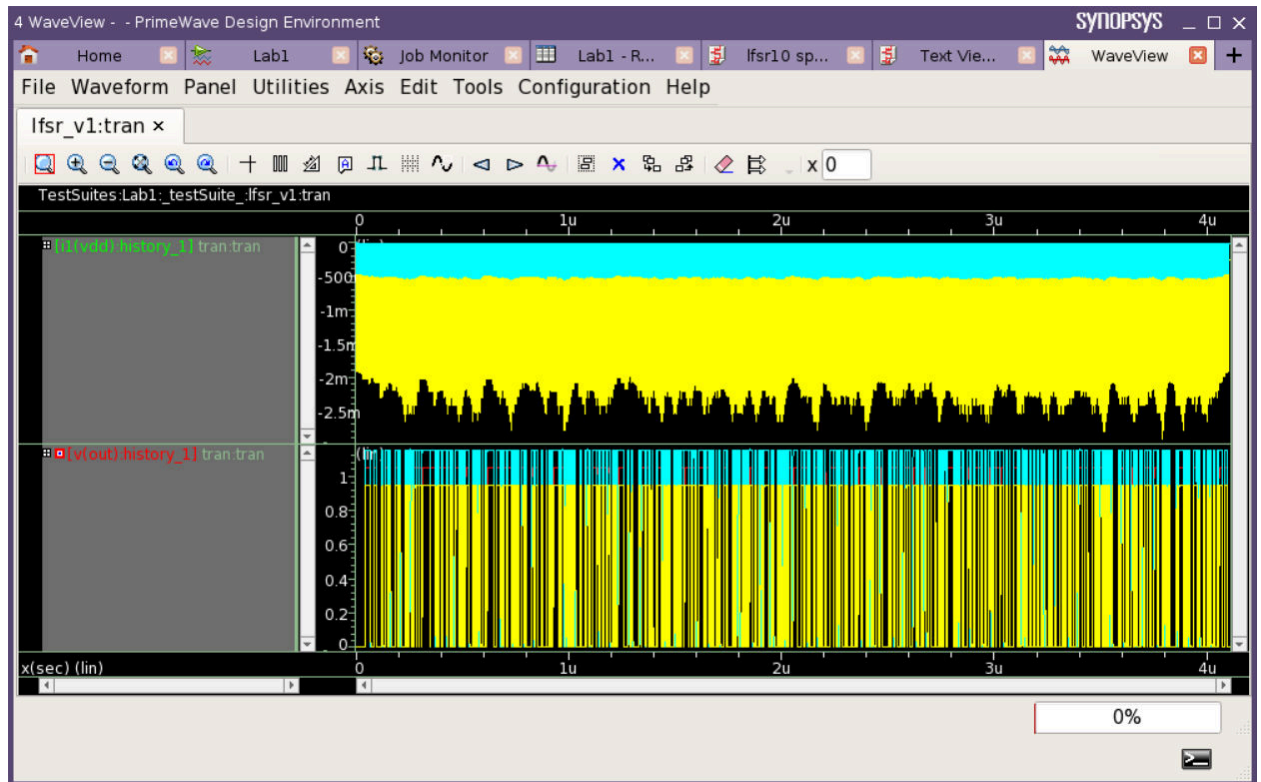
Running a Simulation and Viewing the Results

The screenshot shows the PrimeWave Design Environment ResultsView window. The left sidebar displays the Testbenches tree with 'lfsr_v1' selected. The main area shows a table of simulation results for the 'lfsr_v1' testbench. The table includes columns for Testbench, Equations, Specification, Violations, Min, Max, and Corner values (NOMINAL, fast_pvt, slow_pvt). The results are filtered to show only the 'lfsr_v1' testbench. The table shows that the simulation passed all specifications with 0 violations.

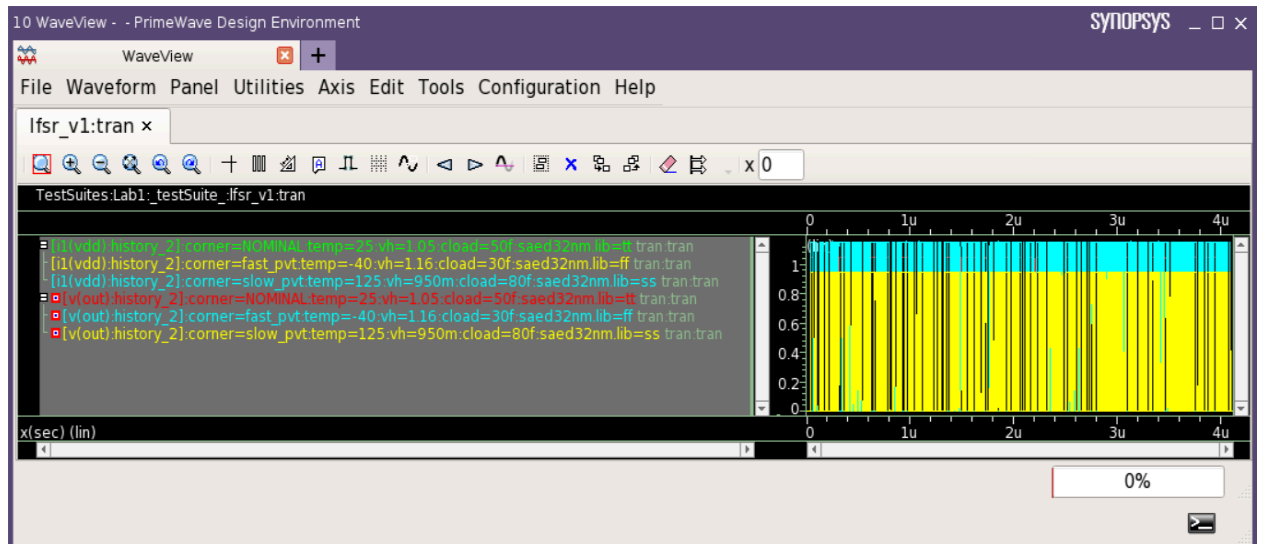
Testbench	Equations	Specification	Violations	Min	Max	Corner NOMINAL	Corner fast_pvt	Corner slow_pvt
lfsr_v1	prms.tranMe...	<0.5	0	62.357u	452.17u	105.54u	452.17u	62.357u
lfsr_v1	clk2out.tran...	<0.6	0	99.359p	513.35p	226.55p	99.359p	513.35p

Buttons at the bottom include 'Re-evaluate', 'Plotting Mode: Append', 'Plot', 'Clear', and 'Calculation done'.

2. Switch to the **WaveView** tab and ungroup the signals to better visualize the analog waveforms.



3. Click the + beside each waveform to observe the details of each corner (corner name and corner parameters).



4. Close WaveView.

See Also

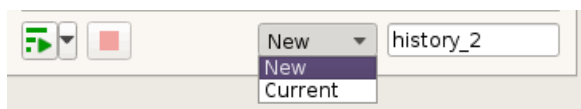
- [Using the ResultsView](#)

Rerunning a Simulation and Viewing History Points

If you make changes to the testbench, such as adding corners or editing specifications, you can choose **Netlist and Run** again to rerun the simulation.

As soon as simulation completes, the next history_point is created by default as history_2. You can choose to run in this new history point or revert to the current one using the drop-down menu.

The previous run is marked as a separate history point. Any new simulation with changes to the existing testbench creates a new history point. In the ResultsView **Testbenches** table the current run is marked in bold.



1. Select the lfsr_v1 testbench in the **Testbenches (history_2)** section of the ResultsView.
2. Observe the results for the prms and clk2Out measurements of each corner.

8 Lab1 - ResultsView - PrimeWave Design Environment

synopsys

Home

Lab1

Job Monitor

Lab1 - ResultsView

File History Testbench Table Help

Testbenches

Description

Testbenches (history_1)

lfsr_v1

Testbenches (history_2)

lfsr_v1

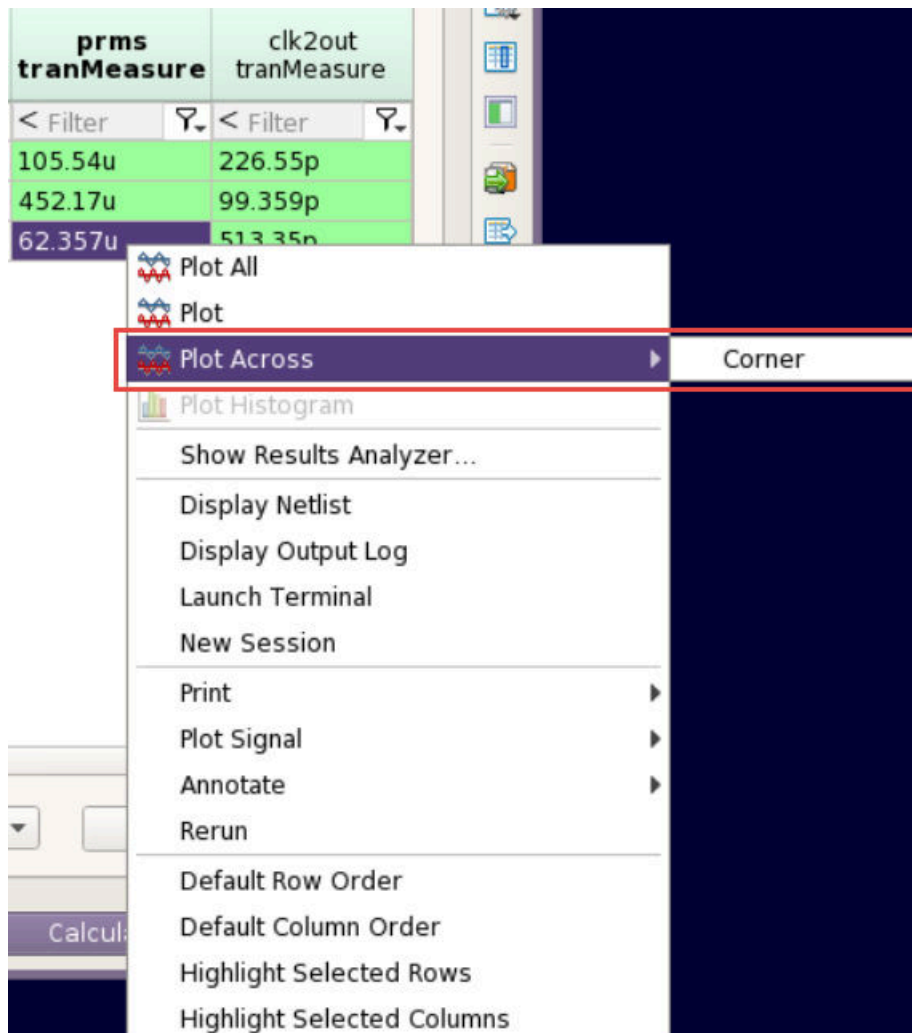
Charts

Summaries

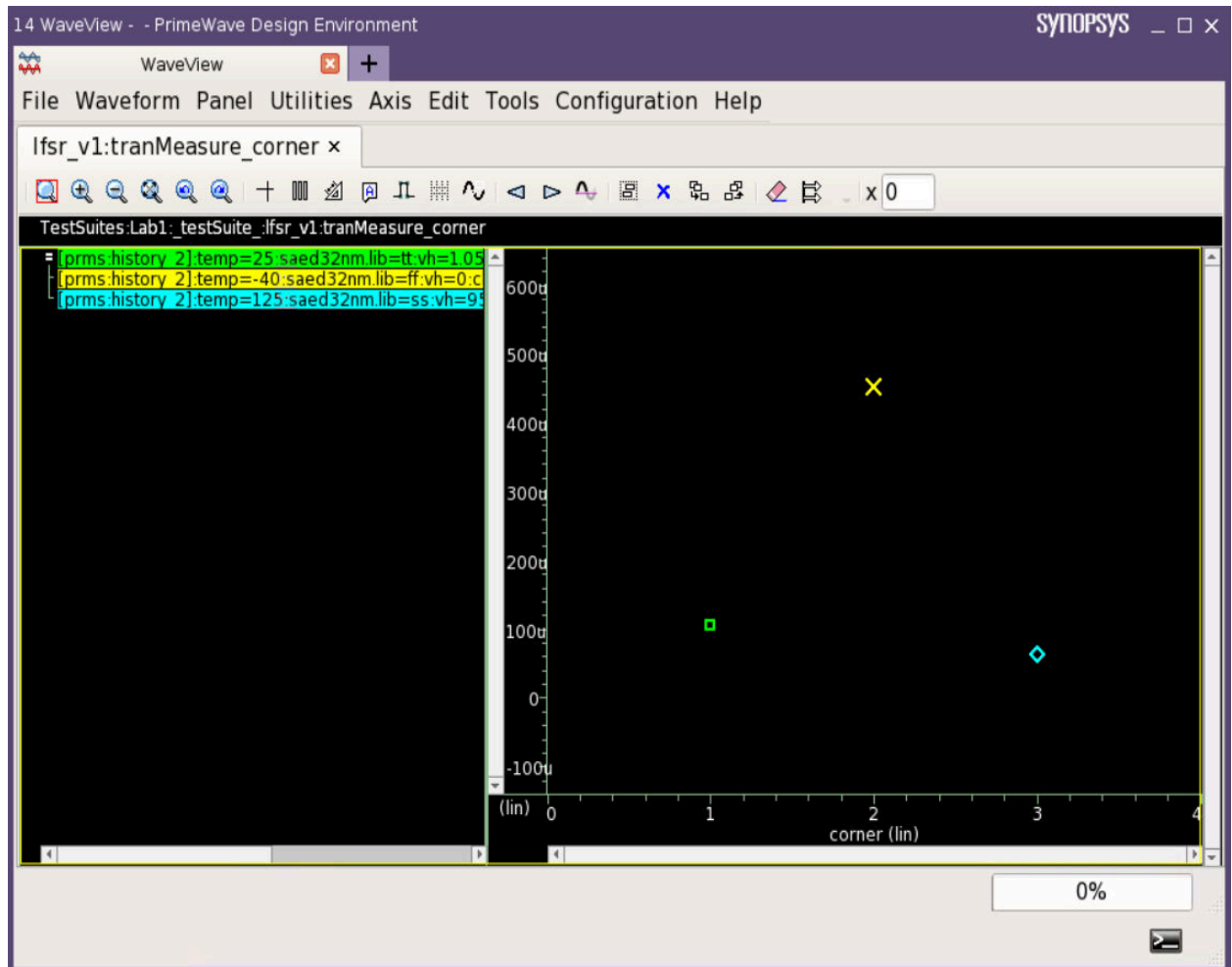
Measurements

er le	Corner Variable cloud	i1(vdd) tran	v(out) tran	prms tranMeasure	clk2out tranMeasure
50f				105.54u	226.55p
30f				452.17u	99.359p
80f				62.357u	513.35p

3. Right-click any prms result to open the menu.
4. Choose **Plot Across > Corner**.



The corners are plotted in the WaveView tool, which opens in a new window.



5. Close WaveView, the ResultsView, and the Job Monitor.

Importing a Post-Layout Version of the Circuit Under Test

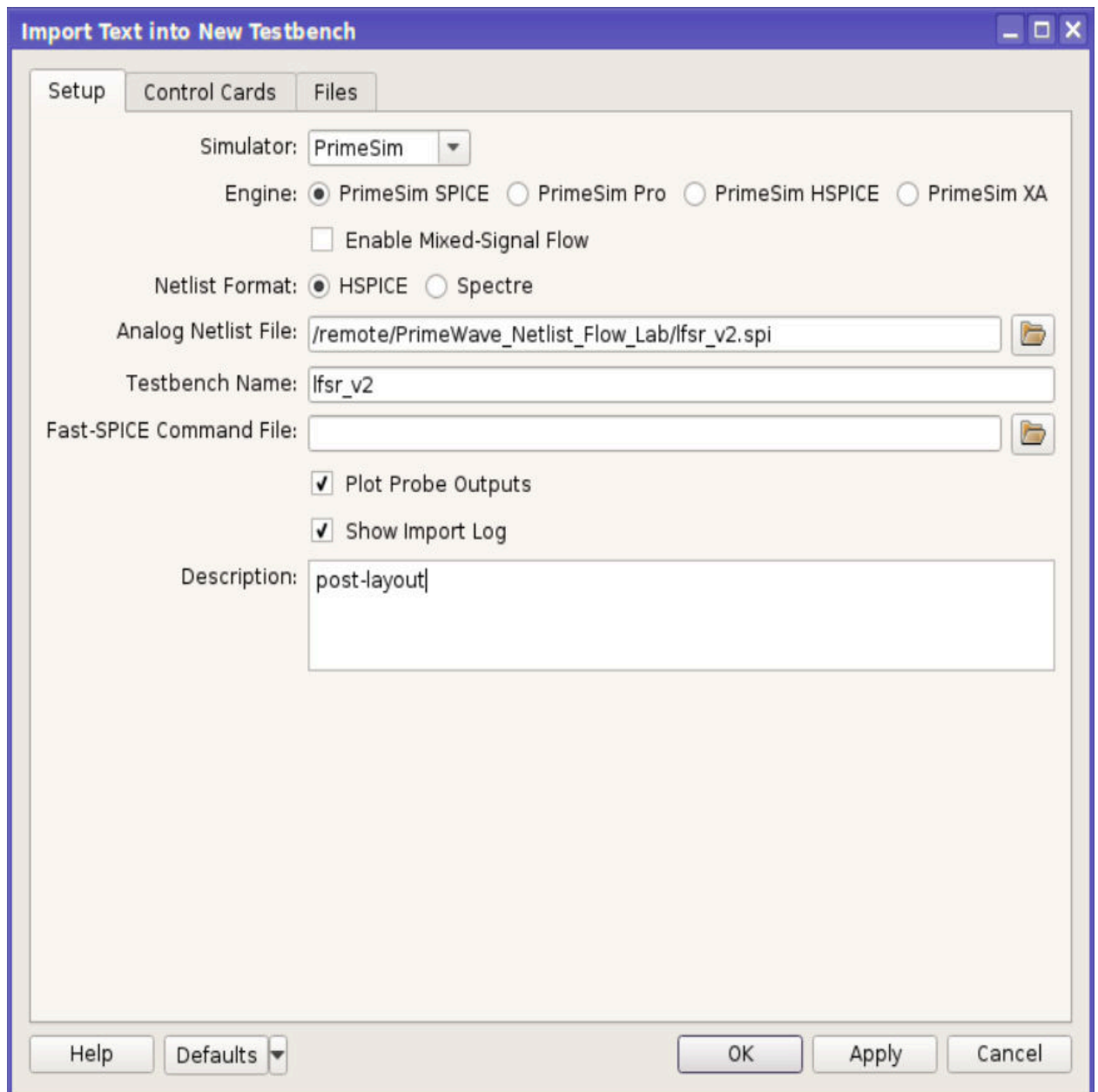
In the previous sections you worked with a testbench built around an ideal or pre-layout version of the circuit under test. It is common practice to create complete testbench netlists around a post-layout version of the circuit under test. In this task, you:

- Import a post-layout version of the 10-bit LFSR
- Use the **Load State** feature to load the specifications from an existing state (rather than manually creating the specifications)

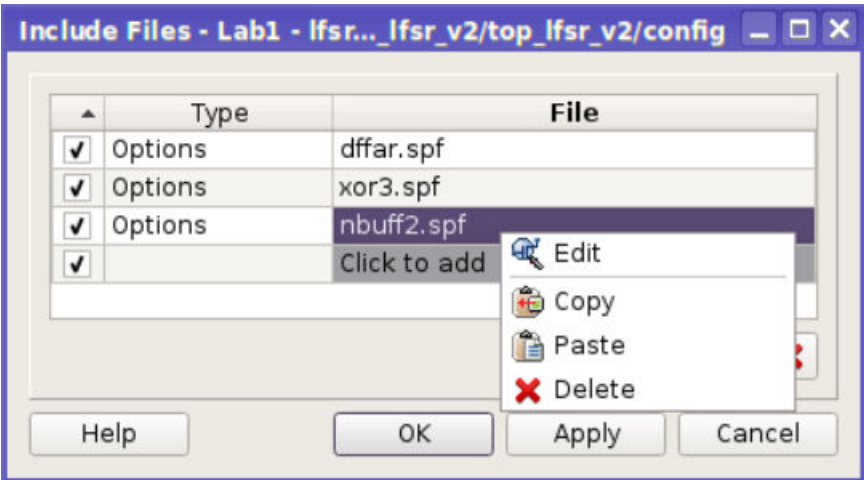
To import the `lfsr_v2.spi` netlist:

1. In the main PrimeWave Design Environment window, choose **Testbench > Add Testbench from Text Import**.
2. In the **Import Text into New Testbench** dialog box that opens, choose **PrimeSim** for the **Simulator** and **PrimeSim SPICE** for the **Engine**.
3. Set the **Analog Netlist File** by browsing to the `lfsr_v2.spi` netlist.
4. Enter **post-layout** for the **Description**.

The dialog box should look as follows:



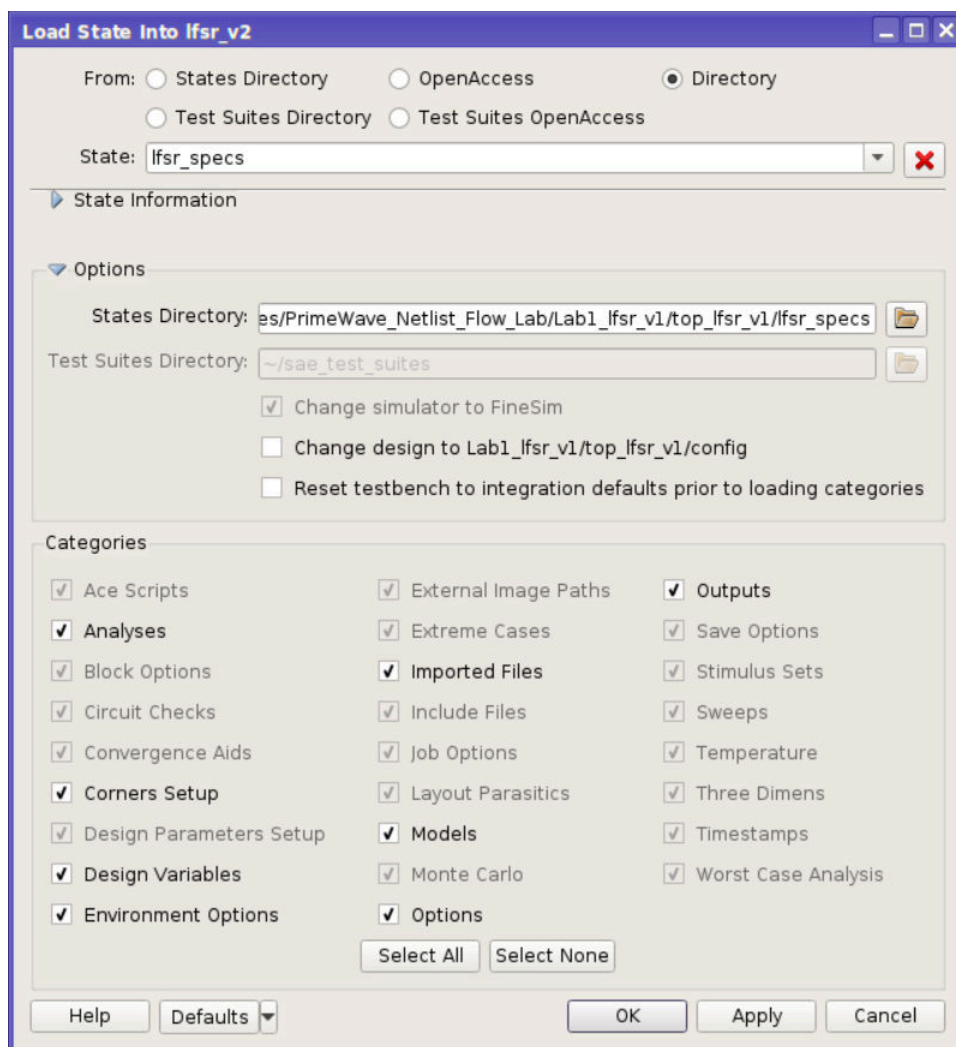
5. Click **OK** to import the file and close the dialog box.
6. Choose **Setup > Include Files** to review the include files. Right-click a file to open the menu and choose **Edit** to view the contents of the .spf file.



- 7. Click **Cancel** to close the **Include Files** dialog box.
- 8. In the **Specifications** table, notice that the default conditions are set.

Outputs		Specifications	Scatter	Histogram	Q-Q	Parametric Reduction	
En	Measurement	Goal Type	Goal/Range Low	Range High	Marginal/Marginal Low	Marginal High	
☒	prms	>		N/A	0%	N/A	
☒	clk2out	>		N/A	0%	N/A	
	Click to add						

- 9. Choose **Testbench > Load STB State** to open the **Load State** dialog box.



10. Choose **From: Directory** and browse to the **lfsr_specs** state.

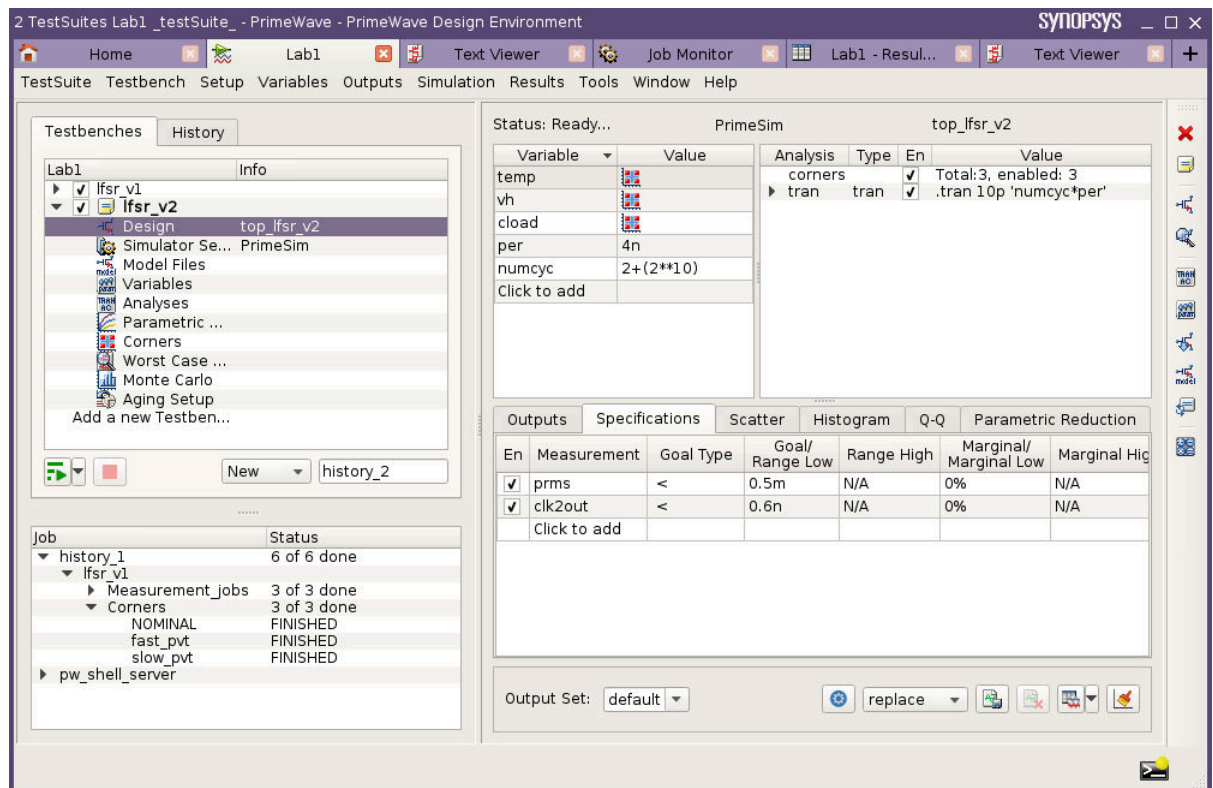
The output state file contains both Output expressions and Specifications goal data.

Note:

The state to be loaded in this tutorial was created by saving the **Outputs** category from the **lfsr_v1** testbench (used in [Importing a Netlist](#)) to an STB state directory. For more information on Saving States, see the *PrimeWave Design Environment User Guide*.

11. Click **OK** to load the state and close the dialog box.

The **lfsr_v2** Specifications table is updated and the lab1 test suite will look like the following screen capture.



12. Make sure both testbenches are enabled and choose **Simulation > Netlist and Run**.

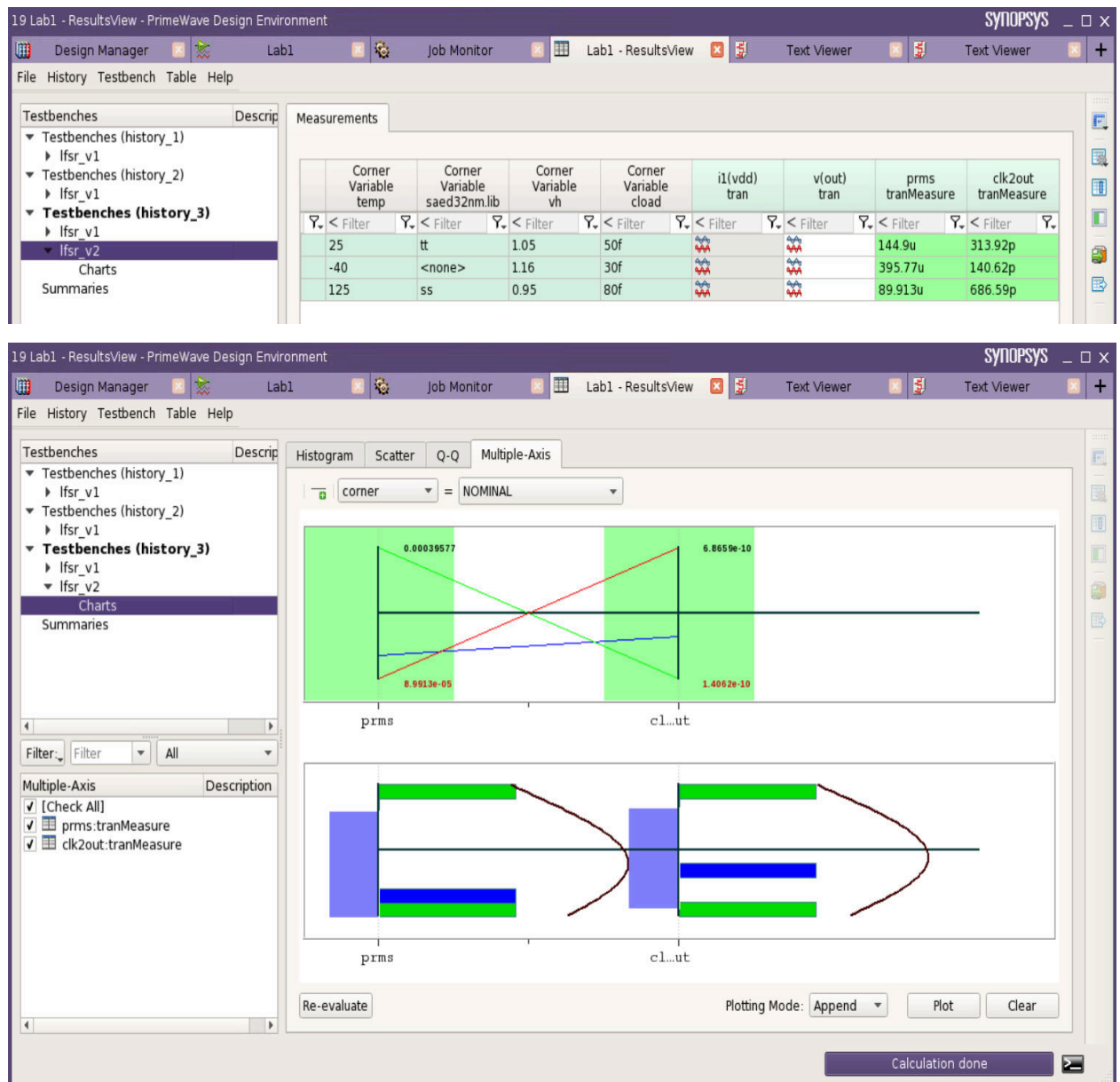
13. Observe the simulation progress in the Job Monitor.

Job	Status
history_1	6 of 6 done
pw_shell_server	
history_2	6 of 6 done
history_3	0 of 6 done (0.0%)
lfsr_v1	
Corners	0 of 3 done (0.0%)
NOMINAL	TRAN (56.9%)
fast_pvt	TRAN (58.7%)
slow_pvt	TRAN (58.0%)
lfsr_v2	
Corners	0 of 3 done (0.0%)
NOMINAL	TRAN (23.9%)
fast_pvt	LOCALLY QUEUED

14. Observe the results in the ResultsView.

Chapter 1: Getting Started With Netlist-Based Flow in the PrimeWave Design Environment

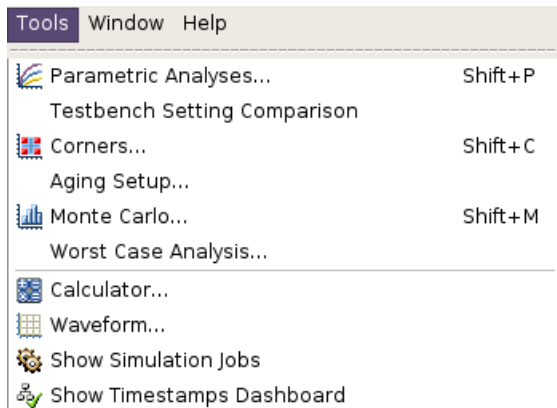
Running an Advanced Analysis: Monte Carlo Analysis



15. Close the ResultsView and the Job Monitor.

Running an Advanced Analysis: Monte Carlo Analysis

The netlist-based PrimeWave Design Environment includes full-featured graphical interfaces for setting advanced analyses like sweeps, corners, MOSRA analysis, and Monte Carlo, available from the **Tools** menu.



Consult the *PrimeWave Design Environment User Guide* for complete instructions on Running Advanced Analyses.

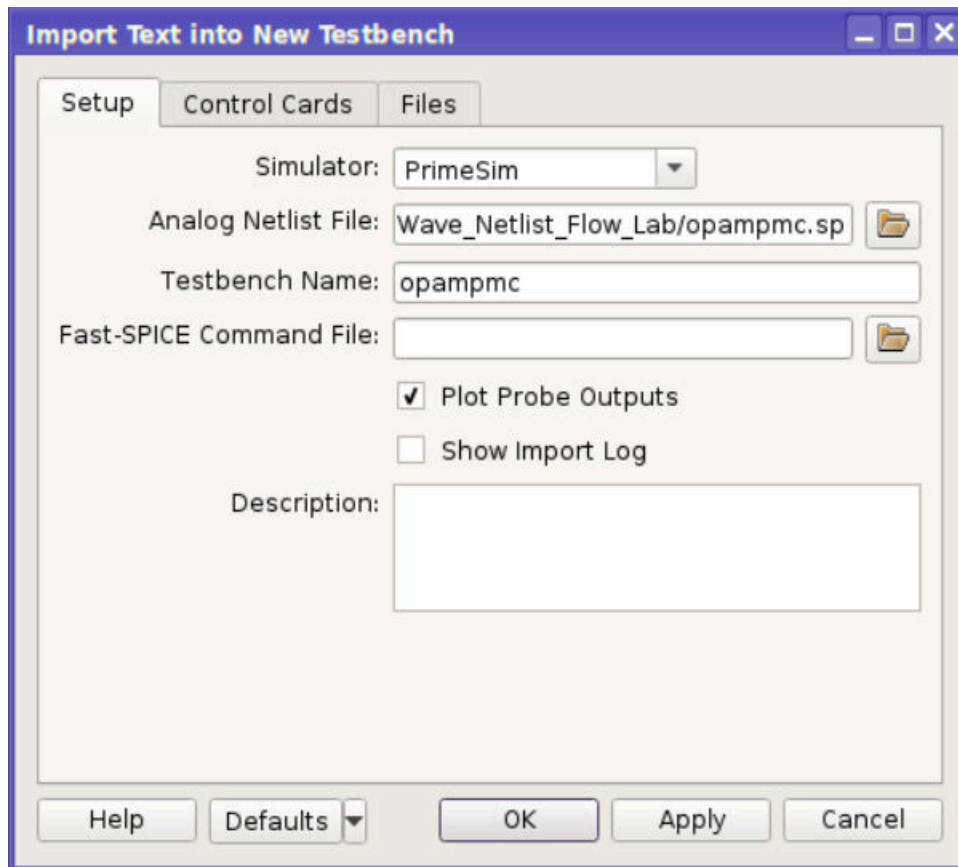
This section describes how to simulate a Monte Carlo analysis and view the updates in the WaveView and the ResultsView tools in the following topics:

- [Adding a New Testbench and Viewing the Model Files](#)
- [Setting Up and Running the Monte Carlo Analysis](#)
- [Viewing the Results](#)
- [Performing Sensitivity Analysis](#)

Adding a New Testbench and Viewing the Model Files

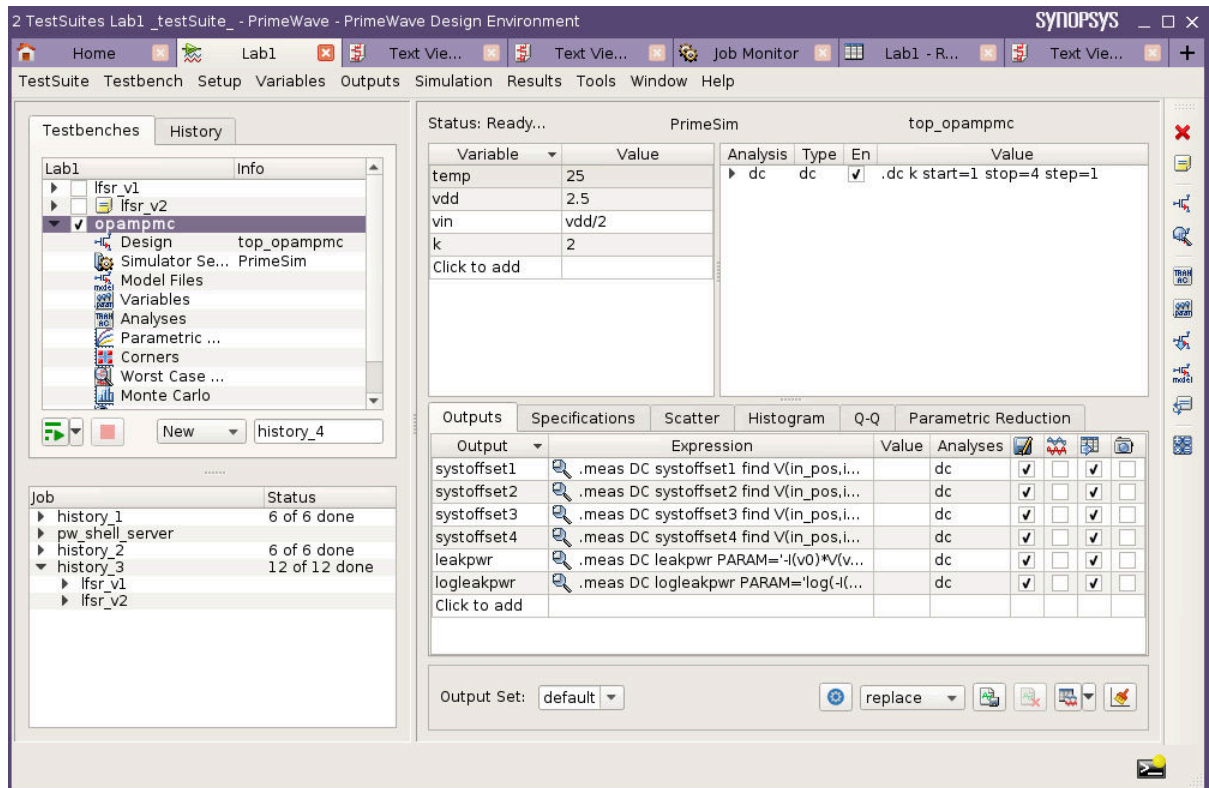
The following procedure demonstrates how to add a new opamp test bench to the existing test suite.

1. Import the analog netlist file `opampmc.sp` following the procedure described in [Importing a Netlist](#).

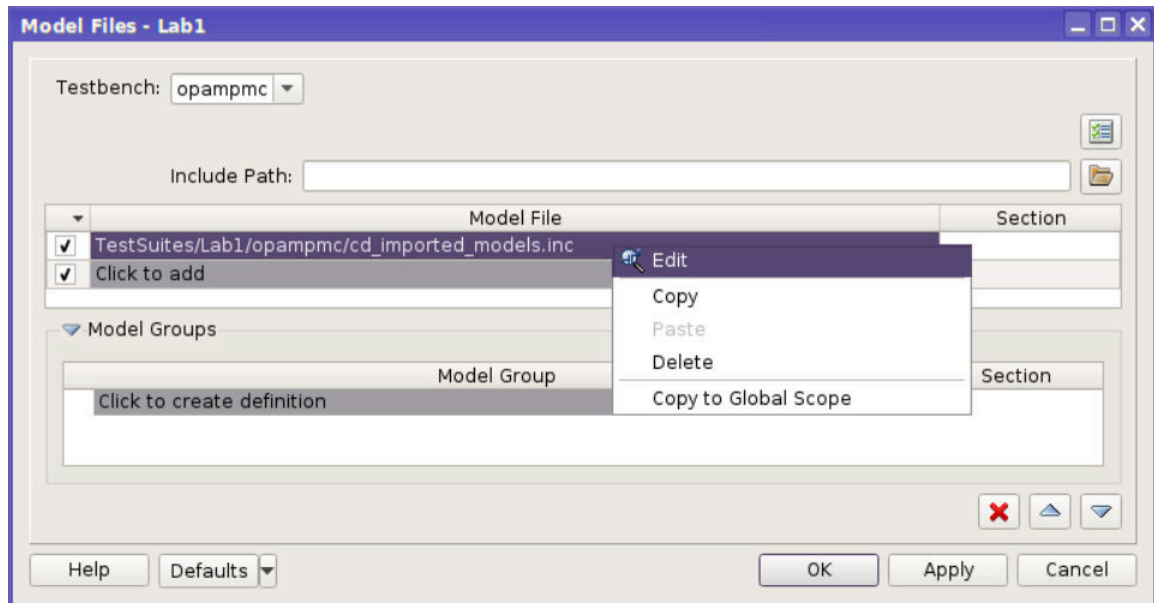


2. Following a successful import of the `opampmc.sp` file, disable the testbenches `lfsr_v1` and `lfsr_v2`.

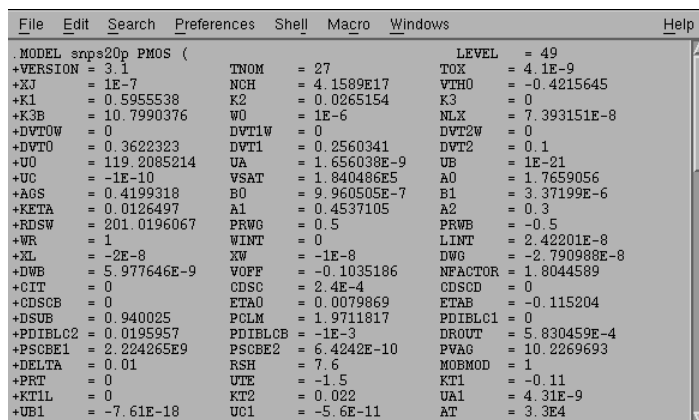
The PrimeWave Design Environment session should now look like the following screen capture.



3. Choose **Setup > Model Files** to check the imported model files. This is a file created from the imported models by the PrimeWave Design Environment.



4. To view the contents, right-click on the model file and choose **Edit**.



```

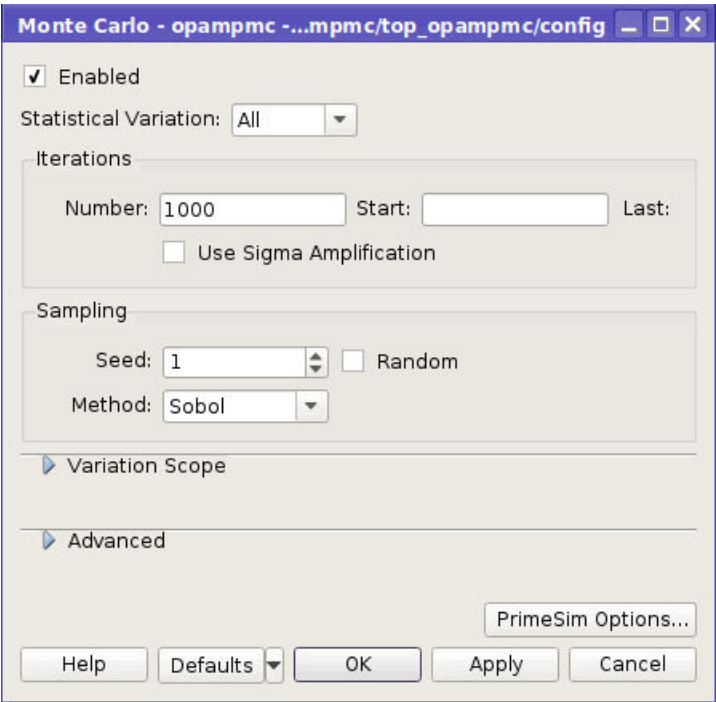
File Edit Search Preferences Shell Macro Windows Help
.MODEL snps20p PMOS (
+VERSION = 3.1
+XJ = 1E-7
+K1 = 0.5955538
+K3B = 10.7990376
+DVTOW = 0
+DVT0 = 0.3622323
+U0 = 119.2085214
+UC = -1E-10
+AGS = 0.4199318
+KETA = 0.0126497
+RDSW = 201.0196067
+WR = 1
+XL = -2E-8
+DWB = 5.977646E-9
+CIT = 0
+CDSCB = 0
+DSUB = 0.940025
+PDIBLC2 = 0.0195957
+PSCBE1 = 2.224265E9
+DELTA = 0.01
+PRT = 0
+KTIL = 0
+UB1 = -7.61E-18
TNOM = 27
NCH = 4.1589E17
K2 = 0.0265154
W0 = 1E-6
DVT1W = 0
DVT1 = 0.2560341
UA = 1.656038E-9
VSAT = 1.840486E5
B0 = 9.960505E-7
A1 = 0.4537105
PRWG = 0.5
WINT = 0
KW = -1E-8
VOFF = -0.1035186
CDSC = 2.4E-4
ETA0 = 0.0079869
PCLM = 1.9711817
PDIBLCB = -1E-3
PSCBE2 = 6.4242E-10
RSH = 7.6
UTE = -1.5
KT2 = 0.022
UC1 = -5.6E-11
LEVEL = 49
TOX = 4.1E-9
VTH0 = -0.4215645
K3 = 0
NLX = 7.393151E-8
DVT2W = 0
DVT2 = 0.1
UB = 1E-21
A0 = 1.7659056
B1 = 3.37199E-6
A2 = 0.3
PRWB = -0.5
LINT = 2.42201E-8
DWG = -2.790988E-8
NFACTOR = 1.8044589
CDSCD = 0
ETAB = -0.115204
PDIBLC1 = 0
DROUT = 5.830459E-4
PVAG = 10.2269693
MOBMOD = 1
KT1 = -0.11
UA1 = 4.31E-9
AT = 3.3E4

```

Setting Up and Running the Monte Carlo Analysis

In this task, you set up the Monte Carlo analysis by varying the transistor threshold voltage, `vth0` and mobility `u0`. These variations are already defined in the netlist and are imported along with model files. (To view the variation models, select **Setup > Include Files**).

1. From the main PrimeWave Design Environment window, choose **Tools > Monte Carlo** to open the Monte Carlo setup dialog box.
2. Fill in the dialog box as follows:
 - Iterations:
 - Number: 1000
 - Start: 1
 - Click **Enabled** to enable the Monte Carlo analysis.



3. Click **OK**. This sets up the Monte Carlo analysis for the opampmc testbench.

The Monte Carlo analysis appears in the main PrimeWave Design Environment window in the Analysis table.

Analysis	Type	En	Value
▶ Monte Carlo	Monte Carlo	✓	Iterations: 1000 Start Iteration: 1
▶ dc	dc	✓	.dc k start=1 stop=4 step=1

4. In the **Outputs** table, enable the **Plot** option for all the measures.

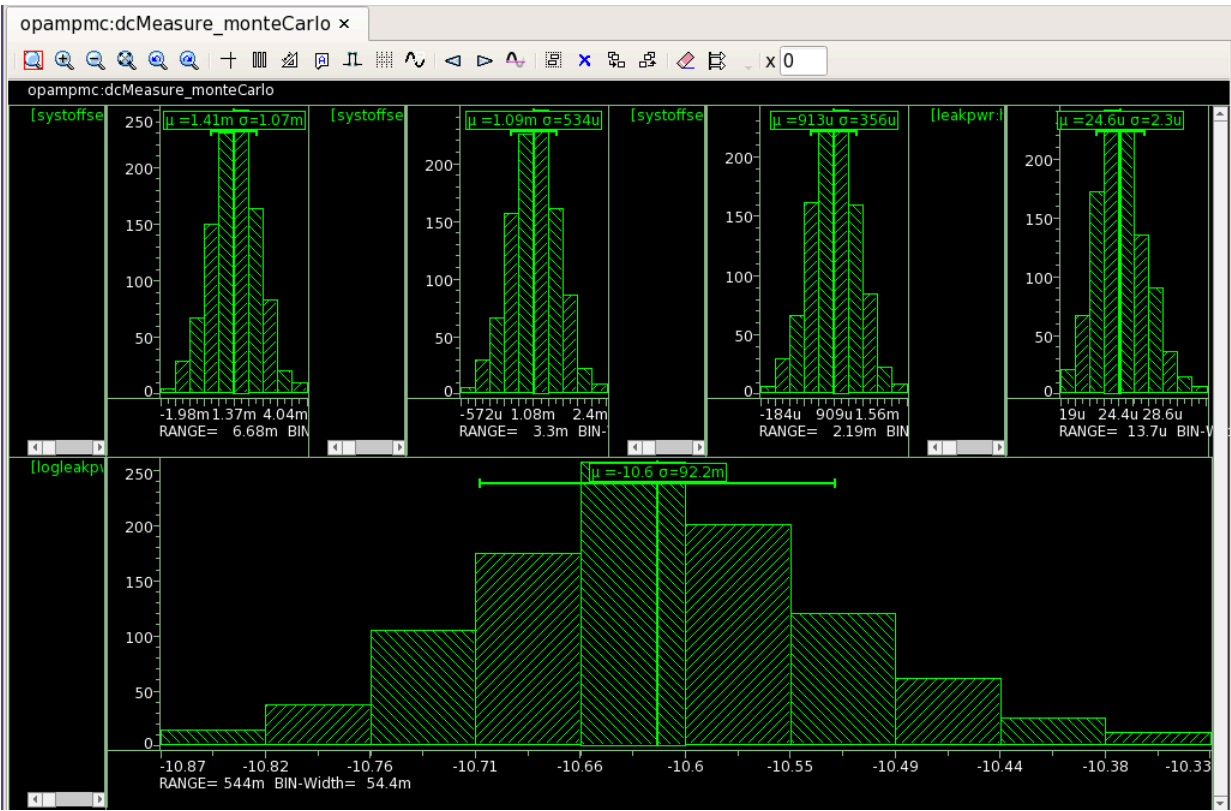
Outputs	Specifications	Scatter	Histogram	Q-Q	Parametric Reduction			
Output	Expression	Value	Analyses			Plot Color		
systoffset1	.meas DC systo...		dc	☒	☒		☒	☐
systoffset2	.meas DC systo...		dc	☒	☒		☒	☐
systoffset3	.meas DC systo...		dc	☒	☒		☒	☐
systoffset4	.meas DC systo...		dc	☒	☒		☒	☐
leakpwr	.meas DC leakp...		dc	☒	☒		☒	☐
logleakpwr	.meas DC logle...		dc	☒	☒		☒	☐
Click to add								

5. Click **Netlist and Run** or choose **Simulation > Netlist and Run** to netlist and run the simulation.

Viewing the Results

After the simulation finishes, the specified outputs are evaluated and plotted in the WaveView tool. The WaveView tool displays statistical histograms for selected measurements.

1. Check the histograms and the mean values of all the measures.



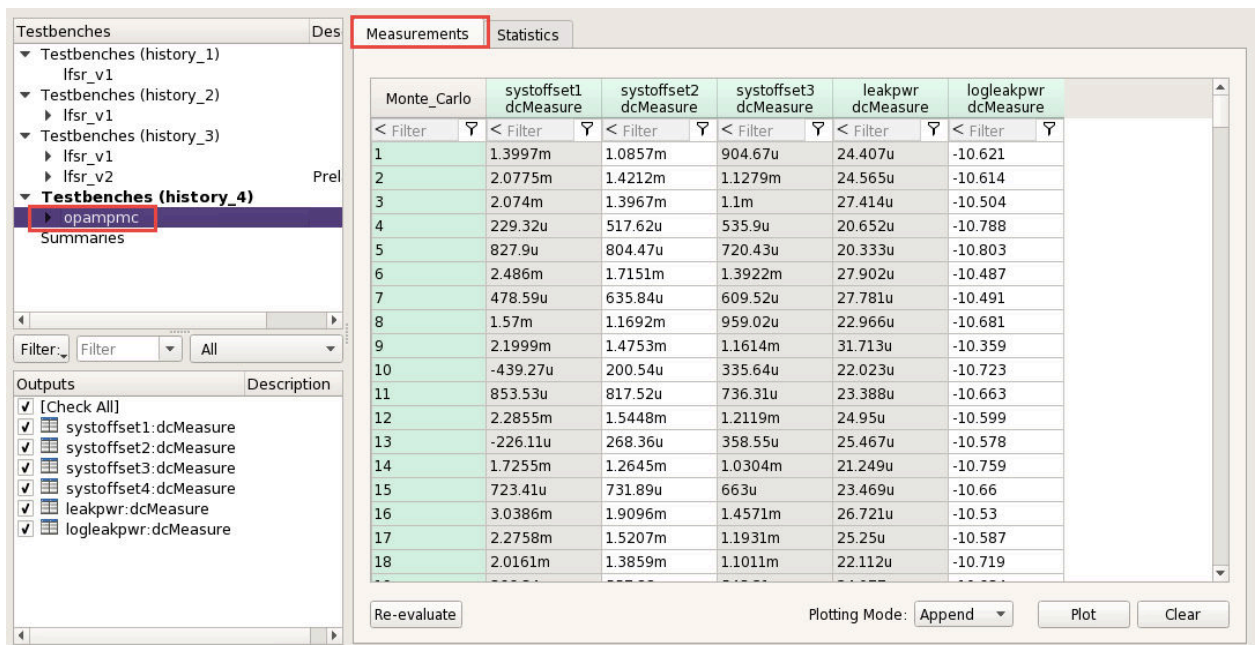
2. Close the WaveView tool.

In the testbench tree of the ResultsView tool, select **Summaries** to see the statistical summary on the **Summary** tab of the results table.

The screenshot shows the ResultsView tool interface. On the left, the testbench tree is expanded to show 'opampmc' and 'Summaries'. On the right, a table displays the statistical summary for various measures across different history iterations.

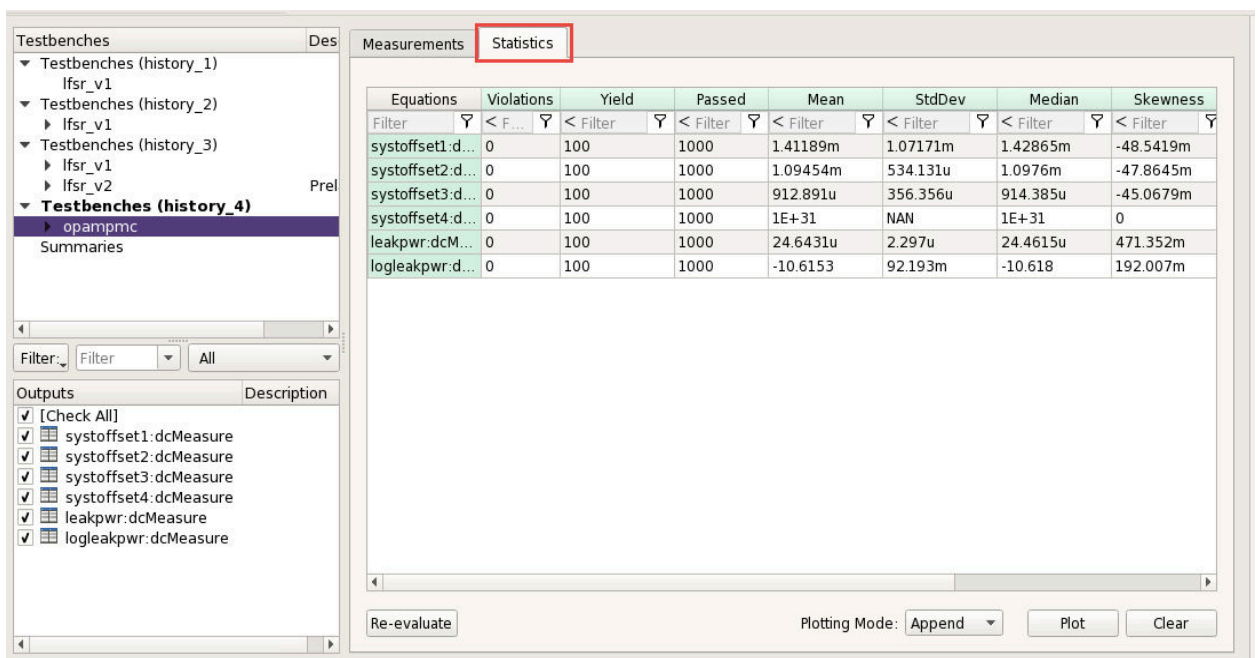
Measure	Unit	Mean	Min	Max	Std Dev	99.99%	99.9%	99%	95%	90%	50%	10%	5%	1%	0.1%
history_2	clk2out.tran...	<0.6n	0	99.359p	513.35p	279.753p	212.06	213.66	213.66	213.66	213.66	213.66	213.66	213.66	213.66
history_3	prms.tranMe...	<0.5m	0	62.357u	452.17u	206.689u	213.66	213.66	213.66	213.66	213.66	213.66	213.66	213.66	213.66
history_3	clk2out.tran...	<0.6n	0	99.359p	513.35p	279.753p	212.06	213.66	213.66	213.66	213.66	213.66	213.66	213.66	213.66
history_3	prms.tranMe...	<0.5m	0	89.913u	395.77u	210.194u	163.04	163.04	163.04	163.04	163.04	163.04	163.04	163.04	163.04
history_3	clk2out.tran...	<0.6n	1	140.62p	686.59p	380.377p	278.96	278.96	278.96	278.96	278.96	278.96	278.96	278.96	278.96
history_4	systoffset1.d...		0	-1.9754m	4.7075m	1.41189m	1.0717	1.0717	1.0717	1.0717	1.0717	1.0717	1.0717	1.0717	1.0717
history_4	systoffset2.d...		0	-572.35u	2.7323m	1.09454m	534.13	534.13	534.13	534.13	534.13	534.13	534.13	534.13	534.13
history_4	systoffset3.d...		0	-184.14u	2.0017m	912.891u	356.31	356.31	356.31	356.31	356.31	356.31	356.31	356.31	356.31
history_4	systoffset4.d...		0	-35.058u	1.5929m	778.646u	267.83	267.83	267.83	267.83	267.83	267.83	267.83	267.83	267.83
history_4	leakpwr:dcM...		0	18.962u	32.661u	24.6431u	2.2976	2.2976	2.2976	2.2976	2.2976	2.2976	2.2976	2.2976	2.2976
history_4	logleakpwr:d...		0	-10.873	-10.329	-10.6153	92.193	92.193	92.193	92.193	92.193	92.193	92.193	92.193	92.193

3. Select the testbench opampmc to see the results for each Monte Carlo iteration on the **Measurements** tab.



Monte_Carlo	systoffset1 dcMeasure	systoffset2 dcMeasure	systoffset3 dcMeasure	leakpwr dcMeasure	logleakpwr dcMeasure
1	1.3997m	1.0857m	904.67u	24.407u	-10.621
2	2.0775m	1.4212m	1.1279m	24.565u	-10.614
3	2.074m	1.3967m	1.1m	27.414u	-10.504
4	229.32u	517.62u	535.9u	20.652u	-10.788
5	827.9u	804.47u	720.43u	20.333u	-10.803
6	2.486m	1.7151m	1.3922m	27.902u	-10.487
7	478.59u	635.84u	609.52u	27.781u	-10.491
8	1.57m	1.1692m	959.02u	22.966u	-10.681
9	2.1999m	1.4753m	1.1614m	31.713u	-10.359
10	-439.27u	200.54u	335.64u	22.023u	-10.723
11	853.53u	817.52u	736.31u	23.388u	-10.663
12	2.2855m	1.5448m	1.2119m	24.95u	-10.599
13	-226.11u	268.36u	358.55u	25.467u	-10.578
14	1.7255m	1.2645m	1.0304m	21.249u	-10.759
15	723.41u	731.89u	663u	23.469u	-10.66
16	3.0386m	1.9096m	1.4571m	26.721u	-10.53
17	2.2758m	1.5207m	1.1931m	25.25u	-10.587
18	2.0161m	1.3859m	1.1011m	22.112u	-10.719

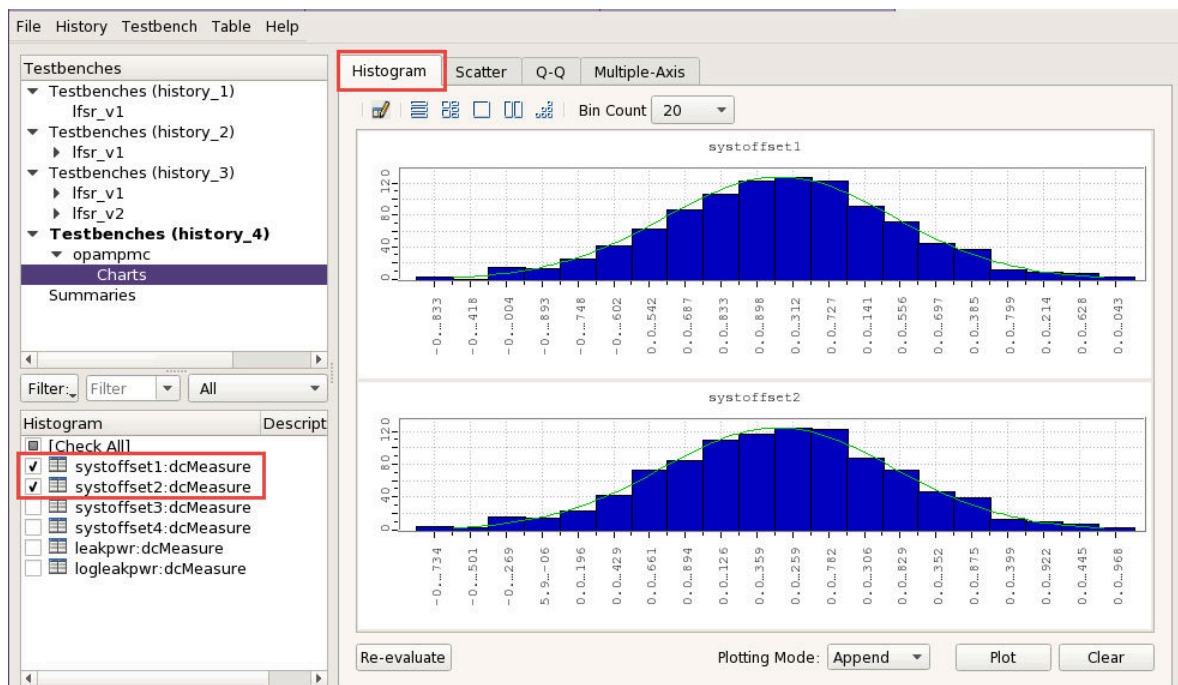
- Switch to the **Statistics** tab in the ResultsView and observe the various statistical quantities, such as partial yield, mean, sigma, skewness, excess kurtosis, and so forth.



Equations	Violations	Yield	Passed	Mean	StdDev	Median	Skewness
systoffset1.d...	0	100	1000	1.41189m	1.07171m	1.42865m	-48.5419m
systoffset2.d...	0	100	1000	1.09454m	534.131u	1.0976m	-47.8645m
systoffset3.d...	0	100	1000	912.891u	356.356u	914.385u	-45.0679m
systoffset4.d...	0	100	1000	1E+31	NAN	1E+31	0
leakpwr:dcM...	0	100	1000	24.6431u	2.297u	24.4615u	471.352m
logleakpwr:d...	0	100	1000	-10.6153	92.193m	-10.618	192.007m

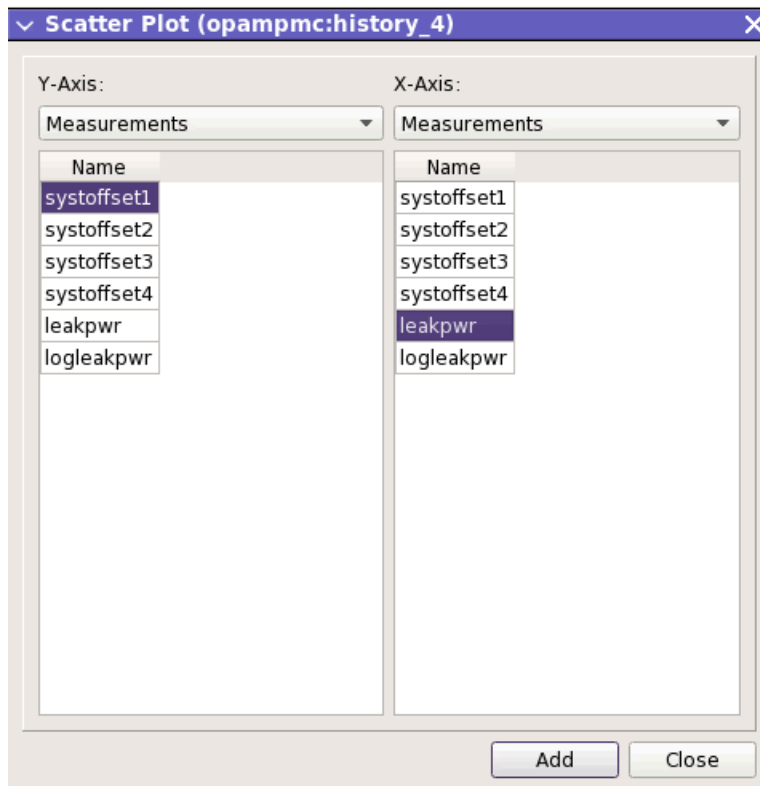
- In the testbench tree, click the arrow next to the opampmc testbench to expand it. Click **Charts** under the opampmc testbench to open the

6. In the bottom-left panel, all the measures are listed. Enable the measures for which histograms are required.

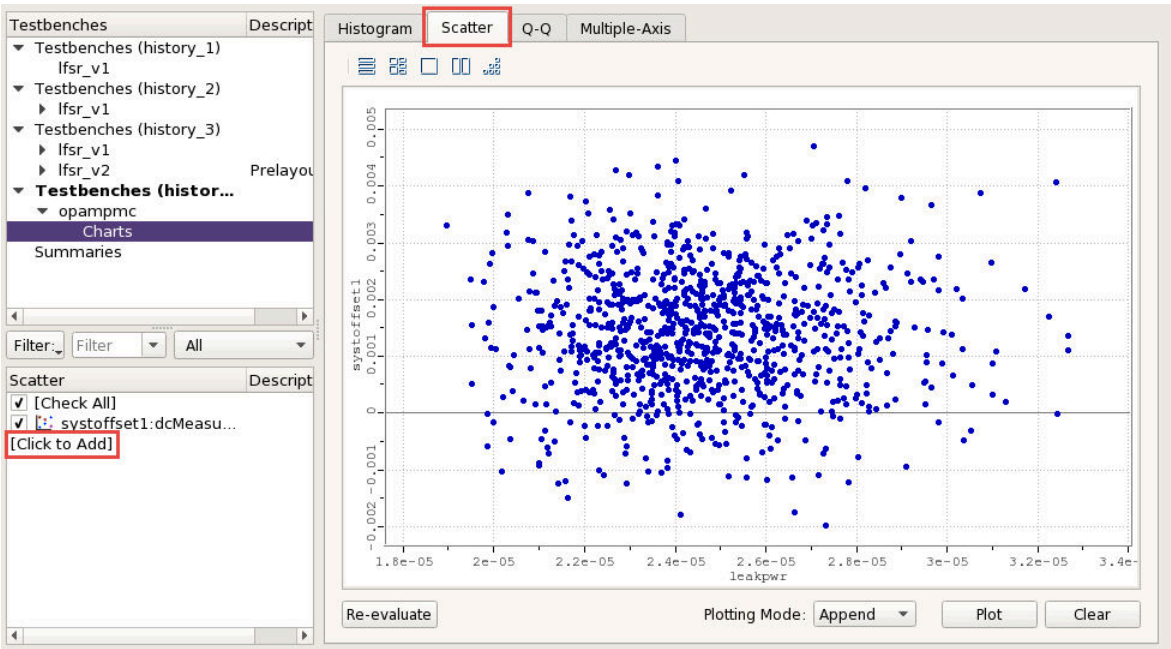


Measurement values inside the green line are within the given specification, and values inside the red region are values in violation of the given specification.

7. Switch to the **Scatter** tab.
8. Click **Click to Add** in the bottom-left panel to open the **Scatter Plot** dialog box.



9. Select any two measures as the x-axis and y-axis and click **Add** to create the scatter plot.

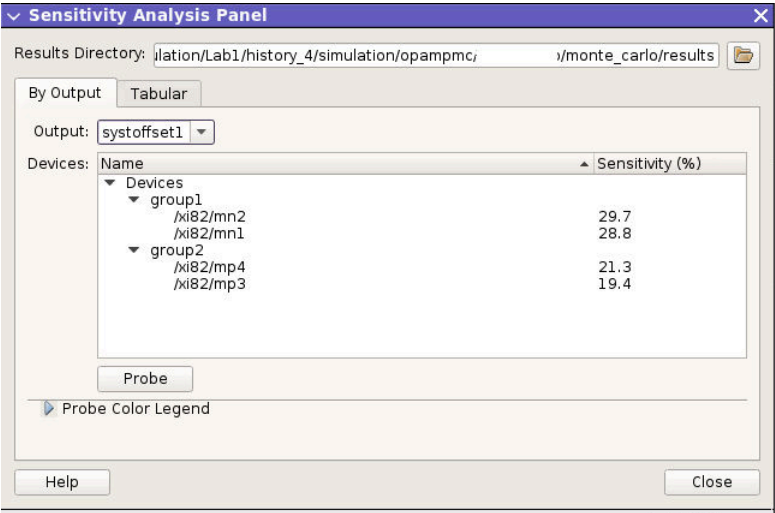


10. Close the ResultsView tool.

Performing Sensitivity Analysis

1. In the main PrimeWave Design Environment window, choose **Results > Data Mining > Analyze Results**.

The **Sensitivity Analysis Panel** opens and displays the sensitivity of each measure on the devices in the netlist.



2. Close the **Sensitivity Analysis Panel**.
3. Save the test suite (**TestSuite > Save**) and exit the PrimeWave Design Environment.

Managing Designs with the Design Manager

This section describes the following tasks in the Design Manager:

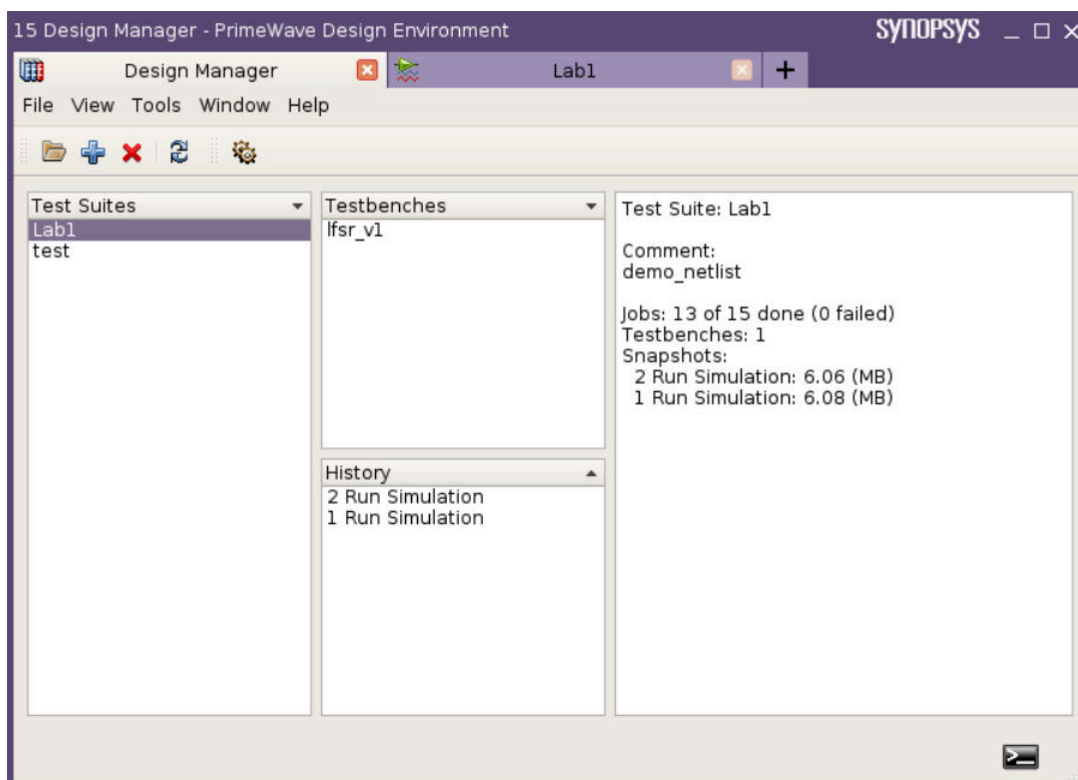
- [Opening the Design Manager](#)
- [Opening Existing Test Suites](#)
- [Deleting Test Suites](#)
- [Opening Results from Previous Simulations](#)
- [Viewing Designs](#)

Opening the Design Manager

To open the Design Manager, click the **Design Manager** button on the **Home** tab.



The Design Manager window opens. Any existing test suites, related testbenches, and run history are displayed.



A list of test suites is shown in the **Test Suites** pane. The PrimeWave Design Environment supports multiple test suites. The **Testbenches** pane lists all testbenches in the selected test suite. The **History** pane shows all history points. The text in the information pane on the far right varies depending on which item is selected in the other panes, but generally includes disk space usage of the simulation results.

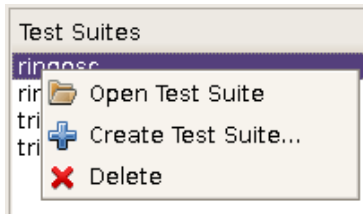
A menu is available from each pane by right-clicking on items in the pane, as illustrated in the sections below.

Opening Existing Test Suites

To open an existing test suite, double-click the name of a test suite you want to open in the **Test Suites** pane of the Design Manager. The test suite opens in the SAE window with any saved testbenches and associated testbench setup.

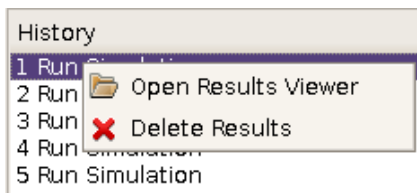
Deleting Test Suites

To delete a test suite, right-click the name of a test suite in the **Test Suites** pane of the Design Manager and choose **Delete** from the menu.



Opening Results from Previous Simulations

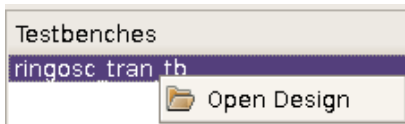
To open previous simulation results, right-click a history point in the **History** pane of the Design Manager and choose **Open Results Viewer** from the menu.



The Results Viewer opens with results from that simulation history point. If you do not yet see any history points, you need to set up and run a simulation first.

Viewing Designs

To view a design in the Design Manager, right-click a design name in the **Testbenches** pane and choose **Open Design** from the menu.



Your design opens in the Text Viewer.

2

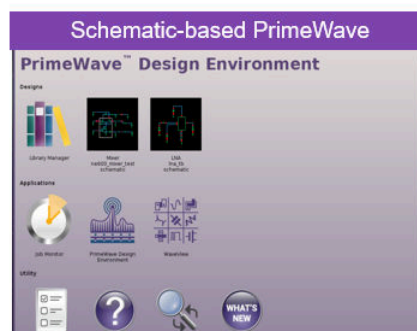
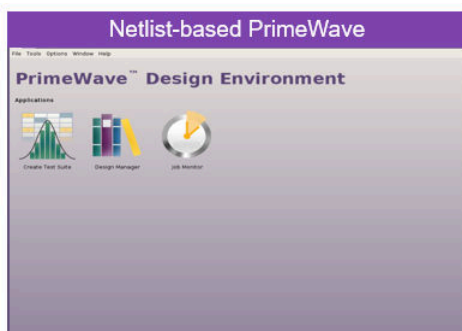
Getting Started With Schematic-Based Flow in the PrimeWave Design Environment

This chapter explains how to prepare your environment to use the schematic-based flow in the PrimeWave Design Environment.

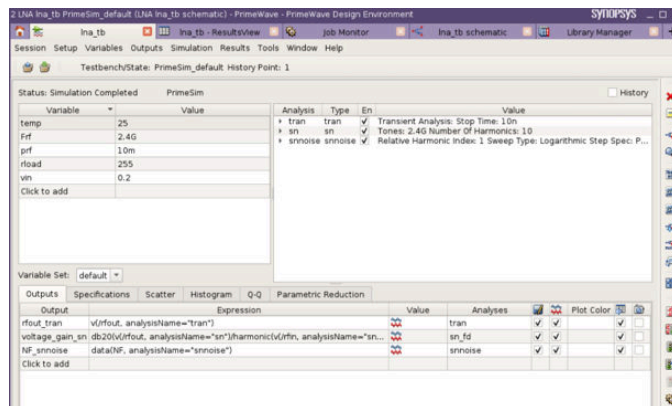
PrimeWave provides two use-models. Use different command line options to invoke:

- Netlist-based use-model (by default): `primewave &`
- Schematic based use-model: `primewave -designType schematic &`

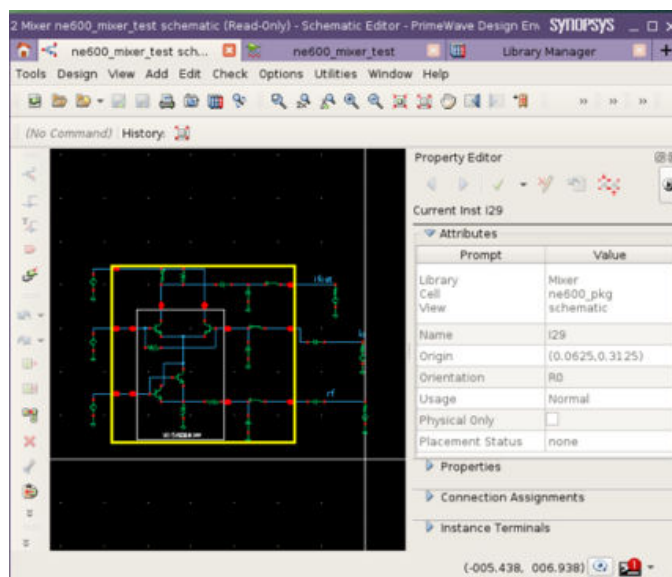
PrimeWave netlist based use-model consist of **Create Test Suite**, **Design Manager** and **Job Monitor**. PrimeWave schematic based use-model consist of **Library Manager**, **Schematic Editor (read-only)**, **PrimeWave Design Environment**, **WaveView** and **Job Monitor**.



Netlist format changes are not required nor recommended. If the original schematic is in Spectre format, you must retain the same format.



PrimeWave opens Synopsys Schematic Editor in read-only mode. PrimeWave uses Synopsys Schematic Editor to setup simulation and post process results.



Quick Start Instructions

The following are the quick start instructions:

Requirements:

- Need PrimeWave version V-2023.12 or above
- Schematics are OA based
- Front-end iPDK (no layout view needed)

- Create lib.defs in work directory
- Replace 3rd party's analogLib with PrimeWave built-in analogLib and include all built-in libraries

```
INCLUDE $SYNOPSYS_CUSTOM_INSTALL/samples/lib.defs
```

Replace analogLib:

```
DEFINE analogLib $SYNOPSYS_CUSTOM_INSTALL/samples/analogLib
```

Replace 3rd party's PDK with Synopsys iPDK (only need front-end iPDK) and retain the design libraries.

Advantages of Standalone PrimeWave

The following are the advantages of standalone PrimeWave:

- Light-weight, focus on simulation.
 - Full capability of PrimeWave. Everything you need for simulation
 - Enables schematic and layout viewing in read-only mode

Any other Custom Compiler feature requires the appropriate license
- Launched in original work directory. There is no need to copy files to different directories.
- If original schematic is in Spectre format, no need to migrate to HSPICE format.
 - Synopsys analogLib has good compatibility with 3rd party's analogLib
 - PrimeWave uses OA based schematics directly
 - PrimeSim can read Spectre format netlists and get compatible results with 3rd party's simulator